

Supplemental Online Content

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This supplemental material has been provided by the authors to give readers additional information about their work.

eAppendix. Supplemental administrative information and methods

SECTION 1: BedMed Committees, Co-investigators, and Staff

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Alberta: Scott Garrison

Saskatchewan: Cathy MacLean

Manitoba: Alexander Singer

Ontario: Braden O'Neill

Participating Medical Clinics

(rand = number of resulting randomizations, provider = number of family physicians or nurse practitioners recruiting at that location)

ALBERTA

Banff: Bear Street Family Physicians (5 providers, 9 rand)

Beaumont: Beaumont Medical (7 providers, 21 rand)

Calgary: Aspen Medical Clinic (1 provider, 16 rand), Associate Clinic (3 provider, 21 rand), Britannia Medical Clinic (1 provider, 3 rand), Crowfoot Village Family Practice (1 provider, 5 rand), Cura Family Practice (3 provider, 27 rand), Dr. Cathryn Kuzyk (1 provider, 10 rand), Dr. Margaret Churcher Professional Corporation (1 provider, 1 rand), East Calgary Health Centre (11 providers,

23 rand), Family Health Clinic (1 provider, 10 rand), Foothills Primary Care Centre (16 providers, 62 rand), Santé Family Medicine (5 providers, 5 rand), Sheldon M Chumir Health Centre (25 providers, 118 rand), South Health Campus (9 providers, 36 rand), Southland Sports Medicine (1 provider, 2 rand), Sunridge Family Medicine Teaching Centre (12 providers, 42 rand), Symons Valley First Care (7 providers, 49 rand), Valley Ridge Family Physicians (2 providers, 14 rand), Valley View Family Practice (1 provider, 2 rand)

Camrose: Smith Clinic (1 provider, 34 rand)

Coaldale: Campbell Clinic Coaldale (1 provider, 2 rand)

Coronation: Coronation Medical Clinic (1 provider, 3 rand)

Drumheller: Riverside Medical Clinic (1 provider, 5 rand)

Edmonton: Alliant Medical Clinic (1 provider, 9 rand), Bearspaw Medical Clinic (1 provider, 1 rand), Capilano Medical Centre (1 provider, 7 rand), Ermineskin Medical Clinic (4 providers, 33 rand), Glenora Medical Clinic (4 providers, 21 rand), Glenwood Medical Clinic (1 provider, 4 rand), Granville Medical Clinic (1 provider, 6 rand), Grey Nuns Family Medicine Centre (7 providers, 34 rand), Heritage Medical Clinic (3 providers, 34 rand), Heritage Village Medicentre (4 providers, 3 rand), Hope Medical Clinic (2 providers, 7 rand), Horizon Family Practice (1 provider, 7 rand), Justik Medical Clinic (2 providers, 16 rand), Kaye Edmonton Clinic (11 providers, 85 rand), Links Clinic (1 provider, 5 rand), Lynnwood Centre (1 provider, 23 rand), MacEwan University Health Centre (5 providers, 6 rand), Magrath Medical Clinic (1 provider, 33 rand), Millcreek Medical Centre (1 provider, 0 rand), Misericordia Family Medicine Clinic (4 providers, 30 rand), New Foundation Medical Centre (1 provider, 10 rand), Northeast Community Health Centre (7 providers, 16 rand), Nova Medical Clinic (2 providers, 14 rand), Ottewell Medical Clinic (6 providers, 41 rand), Sage Seniors Association (2 providers, 7 rand), Shifa Medical Clinic (5 providers, 6 rand), Southwell Medical Centre (1 provider, 3 rand), Summers Medical Clinic (1 provider, 0 rand), The Allin Clinic (3 providers, 44 rand), Town Centre Family Clinic (1 provider, 9 rand)

Edson: Edson Medical Centre (1 provider, 7 rand)

Hinton: Hillcrest Medical Associates (2 providers, 11 rand)

Innisfail: Innisfail Medical Clinic (2 providers, 13 rand)

Lacombe: Mainstreet Medical Services (3 providers, 28 rand)

Leduc: Griffiths Medical Clinic (3 providers, 21 rand), LA Medical Clinic (2 providers, 13 rand)

Lethbridge: Campbell Clinic South (3 providers, 44 rand), Haig Clinic (2 providers, 29 rand), Legacy Medical Centre (4 providers, 23 rand)

Manning: Manning Medical Clinic (2 providers, 11 rand)

Medicine Hat: Horsman Catherine Dr Professional Corporation (1 provider, 21 rand)

Okotoks: Foothills Family Practice (7 providers, 60 rand)

Peace River: Aspen Mile Medical Inc (3 providers, 1 rand), Peace River Medical Clinic (21 providers, 52 rand),

Ponoka: Battle River Medical Clinic (1 provider, 4 rand)

Red Deer: Hevloc Family Medical Clinic (1 provider, 0 rand), Waskasoo Medical Clinic (3 providers, 31 rand)

Rimbey: Rimbey Medical Clinic (1 provider, 15 rand)

Sherwood Park: Synergy Medical Clinic (11 providers, 159 rand)

Spruce Grove: Westgrove Clinic (12 providers, 57 rand)

St. Albert: Horizon Health Medical Clinic (2 providers, 11 rand), St. Albert Medical Clinic (7 providers, 28 rand)
Strathmore: Valley Medical Clinic (8 providers, 40 rand)
Sundre: Moose and Squirrel Medical Clinic (5 providers, 14 rand)
Sylvan Lake: Sylvan Family Health Centre (9 providers, 51 rand)
Taber: Coulee Medical Clinic (1 provider, 1 rand)
Three Hills: Kneehill Medical Clinic (6 providers, 31 rand)
Vulcan: Vulcan Medical Clinic (2 providers, 18 rand)
Wetaskiwin: Wetaskiwin Family Medical Practice (1 provider, 1 rand)
Whitecourt: Life Medical Clinic (2 providers, 13 rand)

BRITISH COLUMBIA

Chilliwack: Crossroads Family Practice (5 provider, 33 rand), Gaetz Family Practice (1 provider, 7 rand), The Young Clinic (6 providers, 62 rand)
Comox: Comox Medical Clinic (1 provider, 12 rand)
Courtenay: Fifth Street Family Practice (1 provider, 8 rand)
Cranbrook: Fisher Peak Family Practice (1 provider, 4 rand)
Fort St. John: Fort St. John Medical Clinic (1 provider, 5 rand)
Langley: Stephen M. Shore Family Practice Inc (1 provider, 10 rand)
Nanaimo: Medical Arts Centre (1 provider, 7 rand)
New Westminster: Care4You (1 provider, 10 rand), New Westminster Family Practice (3 providers, 30 rand)
Port Coquitlam: Wilson Centre Family Practice (12 providers, 95 rand)
Powell River: Harvie Medical Clinic (1 provider, 5 rand)
Richmond: Dr. Cheryl Nagle (1 provider, 12 rand), Dr. Leslie Cohen (1 provider, 17 rand), Well Health Oval A Inc Medical Clinic (1 provider, 7 rand)
Smithers: Coho Clinic (3 providers, 4 rand)
Vancouver: Forward Care Medical Clinic (2 providers, 10 rand)

SASKATCHEWAN

Beechy: Beechy Primary Care Clinic (1 provider, 2 rand)
Kyle: Kyle Primary Care Clinic (1 provider, 3 rand)
Lucky Lake: Lucky Lake Primary Care Clinic (1 provider, 5 rand)
Regina: 5th Avenue Medical (1 provider, 1 rand), Northgate Medical Centre (1 provider, 0 rand)
Rosetown: Rosetown & District Primary Care Centre (1 provider, 4 rand)
Saskatoon: West Winds Primary Health Centre (17 providers, 34 rand), Willowgrove Medical Group (1 provider, 7 rand)

MANITOBA

Winkler: CW Wiebe Medical Centre (1 provider, 5 rand)

Winnipeg: Access Winnipeg West (9 providers, 43 rand), Atlantic Medical Centre (1 provider, 1 rand), Concordia Health Associates (1 provider, 14 rand), Family Medical Centre (6 providers, 47 rand), Kildonan Medical Centre (7 providers, 24 rand), Meadowood Medical Centre (3 providers, 69 rand), Memorial Medical Clinic Inc (1 provider, 4 rand)

ONTARIO

Brantford: Avenue Medical Centre (1 provider, 0 rand)

Collingwood: Collingwood Medical Centre (1 provider, 10 rand)

Guelph: Dawson Road Family Medical Clinic (1 provider, 10 rand)

Hamilton: Dr. C Prangley Medicine Prof Corp (1 provider, 8 rand), Robert Kerr (1 provider, 4 rand)

North York: Bathurst Family Physicians (3 providers, 5 rand)

Ottawa: Advanced Access Medical Clinics (1 provider, 6 rand)

Richmond Hill: Wise Health Medical Center (1 provider, 2 rand)

Toronto: Duncan Family Docs (2 providers, 5 rand), Platinum Medical FHO (1 provider, 3 rand), Prince Edward Medical (1 provider, 2 rand), Scarborough Health Network (1 provider, 2 rand)

Windsor: DeMarco and Associates (1 provider, 8 rand)

SECTION 2: Administrative Health Data (ICD9/10) Outcome Definitions

Overview: The following describes the definitions used to screen for the occurrence of BedMed outcomes in administrative health data in the 3 provinces (British Columbia, Alberta, Manitoba) from which patient level data was available. Clinically, it is notable that the definitions for glaucoma and dementia differ somewhat in that Manitoba (and not British Columbia / Alberta) included dispensations of diagnosis defining medications in the outcome definition. For glaucoma, this difference is likely immaterial as we only used administrative health claims indications of glaucoma as a flag to seek information from the patient, and the primary care provider, as to whether the patient had this diagnosis. For dementia, the difference is more meaningful as (to avoid potentially upsetting participants) we accepted administrative claims diagnoses of dementia without seeking corroboration from patients or primary care providers. Hence a new diagnosis of dementia was more sensitively picked up in Manitoba, than in other participating provinces. Manitoba also used ICD9 codes to detect fractures as many fractures were believed to be managed in the community, or as outpatients.

ICD9/10 Codes used to detect outcomes in administrative health data

Events	Definitions Used for Alberta Data	Definitions Used for British Columbia Data	Definitions Used for Manitoba Data
Acute Coronary Syndrome (ACS) Hospitalization or ED visit	ICD10: I20%, I21%, I22%, I24%	ICD10: I20%, I21%, I22%, I24%	ICD10: I20%, I21%, I22%, I24%
	Note: ACS/MI Hospitalization/ED visit: qualified by primary diagnosis	Note: ACS/MI Hospitalization/ED visit: qualified by primary diagnosis	Note: ACS/MI Hospitalization/ED visit: qualified by primary diagnosis
Stroke Hospitalization or ED visit	ICD10: H34.1%, I60%, I61%, I63%, I64%	ICD10: H34.1%, I60%, I61%, I63%, I64%	ICD10: H34.1%, I60%, I61%, I63%, I64%
	Note: Stroke Hospitalization/ED visit: qualified by primary diagnosis	Note: Stroke Hospitalization/ED visit: qualified by primary diagnosis	Note: Stroke Hospitalization/ED visit: qualified by primary diagnosis
Heart Failure (HF) Hospitalization or ED visit	ICD10: I50%	ICD10: I50%	ICD10: I50%
	Note: HF Hospitalization/ED visit: qualified by primary diagnosis	Note: HF Hospitalization/ED visit: qualified by primary diagnosis	Note: HF Hospitalization/ED visit: qualified by primary diagnosis

Events	Definitions Used for Alberta Data	Definitions Used for British Columbia Data	Definitions Used for Manitoba Data
Hip Fracture	ICD10: S72.0%, S72.1%, S72.2%	ICD10: S72.0%, S72.1%, S72.2%	ICD10: S72.0%, S72.1%, S72.2% ICD9: 820% Physician Billing Codes: • 0865 fractures, lower extremity, femur, neck, closed reduction, cast or traction • 0868 fractures, lower extremity, femur, neck, open reduction with internal fixation • 0870 fractures, lower extremity, femur, neck, prosthetic replacement • 0872 fractures, lower extremity, intertrochanteric, closed reduction • 0874 fractures, lower extremity, intertrochanteric, open reduction The following must have Dx code 820 on same claim: • 1149 joints, arthroplasty, hip, femoral head replacement type • 1150 joints, arthroplasty, hip (cup or total hip replacement prosthesis) • 1154 joints, arthroplasty, hip, femoral head, where previous prosthesis, etc require removal • 1415 total hip arthroplasty • 1423 bipolar hip arthroplasty
	Event qualified by: 1 in-hosp/ED Dx (ANY Dx, not just primary) or 1 procedure service code	Event qualified by: 1 in-hosp/ED Dx (ANY Dx, not just primary)	Event qualified by: 1 in-hosp/ED Dx (ANY Dx, not just primary) or medical claim for hip fracture repair

Events	Definitions Used for Alberta Data	Definitions Used for British Columbia Data	Definitions Used for Manitoba Data
New Glaucoma (No history)	ICD10: H40.1%, H40.6%, H40.8%, H40.9% ICD9: 365.1%, 365.3%, 365.9%	ICD10: H40.1%, H40.6%, H40.8%, H40.9% ICD9: 365.1%, 365.3%, 365.9%	ICD10: H40.1%, H40.6%, H40.8%, H40.9% ICD9: 365.1%, 365.3%, 365.9% (truncated to 3 digits when necessary if 5-digit code is missing on claim) Glaucoma Surgery Tariff: 5547 IRIS AND CILIARY BODY, SURGICAL TRABECULECTOMY OR SIM FILTER PROC FOR GLAUCOMA ATC codes for Drugs to Treat Glaucoma: S01E (includes all subclasses)
	Event qualified by: 1 claim or 1 in-hosp/NACRS Dx (ANY Dx, not just primary)	Event qualified by: 1 claim or 1 in-hosp/ED Dx (ANY Dx, not just primary)	Event qualified by: 1 claim or 1 in-hosp/ED Dx (ANY Dx, not just primary) or medical claim Dx or billing code for surgery or 1+Rx for antiglaucoma preparations and miotics
New Dementia (No history)	ICD10: F00%, F01%, F03%, G30% ICD9: 290%, 331%	ICD10: F00%, F01%, F03%, G30% ICD9: 290%, 331%	ICD10: F00%, F01%, F03%, G30% ICD9: 290%, 331% ATC codes for Drugs to Treat Dementia: N06D (includes all subclasses)
	Event qualified by: 1 claim or 1 in-hosp/ED Dx (ANY Dx, not just primary)	Event qualified by: 1 claim or 1 in-hosp/ED Dx (ANY Dx, not just primary)	Event qualified by: 1 claim or 1 in-hosp/ED Dx (ANY Dx, not just primary) or medical claim Dx or 1+Rx for anti-dementia drugs

Events	Definitions Used for Alberta Data	Definitions Used for British Columbia Data	Definitions Used for Manitoba Data
Non-Vertebral Fracture	ICD10: S02% S22.2% - S22.9% S32.1% - S32.6% S42%, S52%, S62% S72.0%, S72.1%, S72.2% S72.3% - S72.9% S82%, S92%, T02%, T10%, T12% T14.2% ICD9: N/A	ICD10: S02% S22.2% - S22.9% S32.1% - S32.6% S42%, S52%, S62% S72.0%, S72.1%, S72.2% S72.3% - S72.9% S82%, S92%, T02%, T10%, T12% T14.2% ICD9: N/A	ICD10: S02% S22.2% - S22.9% S32.1% - S32.6% S42%, S52%, S62% S72.0%, S72.1%, S72.2% S72.3% - S72.9% S82%, S92%, T02%, T10%, T12% T14.2% ICD9: 800-804, 806.6, 806.7, 807-807.4, 808, 810-819.1, 820-829.1 (truncated to 3 digits when necessary if 5-digit code is missing on claim)
	Event qualified by: 1 in-hosp/ED Dx (ANY Dx, not just primary)	Event qualified by: 1 in- hosp/ED Dx (ANY Dx, not just primary)	Event qualified by: 1 in- hosp/ED Dx (ANY Dx, not just primary) or medical claim Dx Note that hip fractures are a subset of non-vertebral fractures, and the codes overlap.
New Nursing Home Admission (No history)	Claims Service Location Code = 'NH'	Billing Codes: 00114, 13334, 00115 And Service Location Code='C' (Residential care / assisted living residence)	Admission to Long-term care using MB LTC utilization data: http://mchp-appserv.cpe.umanitoba.ca/dataDescriptions.php?ds=LTCUtilization
Last Medical Service Date	The last date of contact using PIN, Claims, DAD, NACRS/ED, and LAB data	The last date of contact using PharmaNet, MSP, DAD, ED, and Chronic Disease Registry data	The last date of contact using hospitalizations, medical claims, DPIN prescription dispensations, EDIS emergency dept visits, long-term care admission, and lab test data.
	Note: Data coverage up to Nov 30, 2023; except for NACRS/ED, DAD up to Oct 31, 2023.	Note: Data coverage up to Dec 31, 2023; except for DAD up to Sep 30, 2023; and Chronic Disease Registry to Mar 31, 2023.	Note: Data coverage up to March 31, 2023

SECTION 3: Outcome Adjudication

Overview: The Event Adjudication Committee met periodically over the course of the trial to review information gathered on participants and determine whether potential outcome events should be accepted or rejected. For each meeting, the committee had three voting members present, each a practicing family physician (comprised of the trial principal investigator and two of five additional clinicians, depending on availability). A research assistant controlled the presentation of data – which included both self-reported events, and administrative health data. All committee members were fully blinded to allocation. In general, if patient self-report and administrative health data appeared to corroborate each other, events were accepted without further investigation. However, if events appeared questionable, or were present in only one of these sources (e.g. an administrative health data diagnosis of myocardial infarction with no corresponding participant report), then the participant’s family physician was contacted (by fax or phone) for more information, and for their opinion on whether the event of interest actually occurred. This process was followed for hospitalization/ED visit for ACS, stroke, or HF, as well as for hip fracture, non-vertebral fracture, and new glaucoma diagnosis. We did not seek confirmation from primary care providers for 1) death or continuing care admission (believing both of these to be accurately reported by both family and administrative health data), 2) new diagnosis of dementia (believing this to be a sensitive subject from the participant’s perspective and not wanting our enquiry to cause them distress), and 3) all-cause unplanned hospitalization (given the number of hospitalizations/ED visits was high for the trial as a whole and we felt this could overwhelm the primary care providers). Adjudication rules were evolved as cases were encountered and the resulting rules used to resolve discrepancies is reported below.

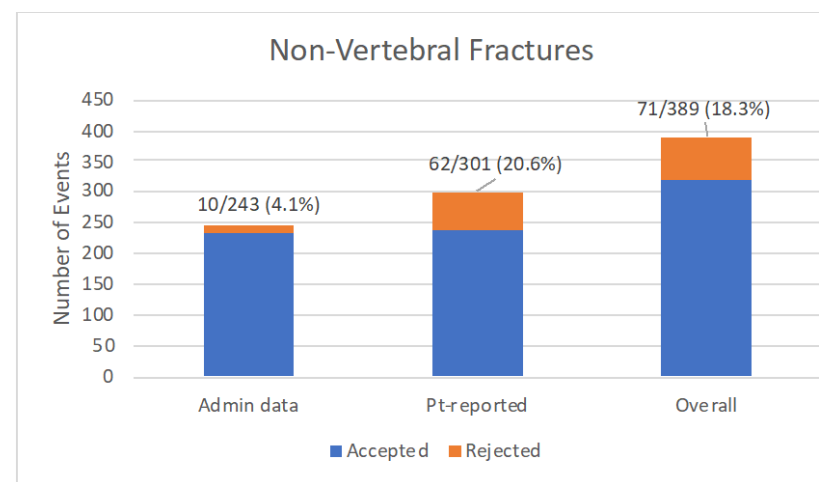
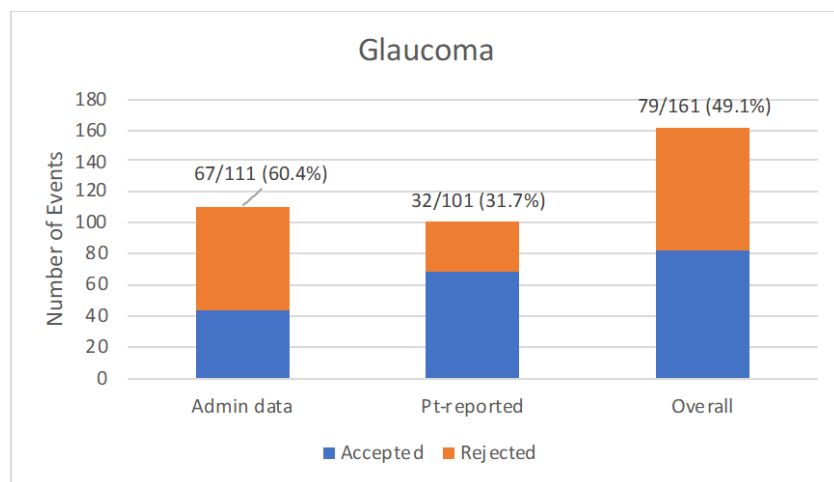
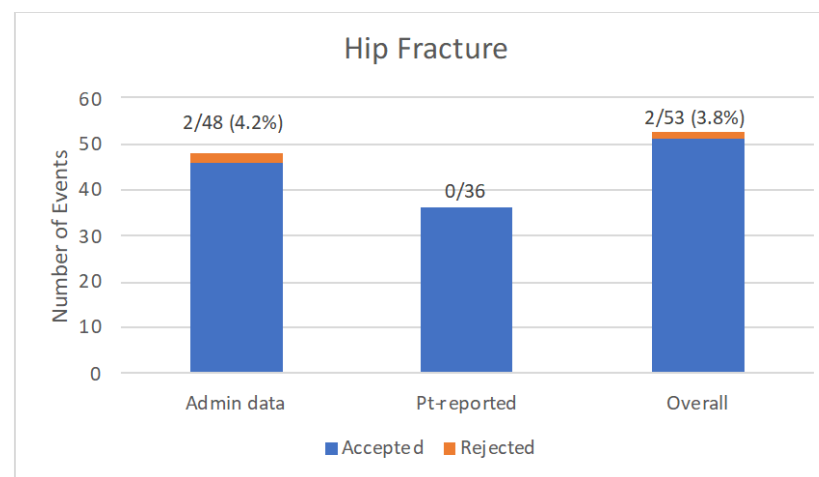
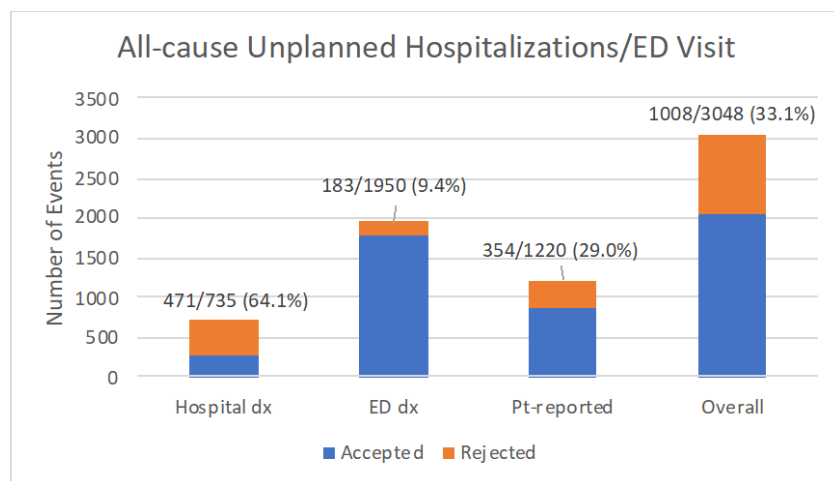
Proportion of Potential Outcomes Disallowed: Overall, a diagnosis provided as the reason for hospitalization within administrative health data was highly likely to indicate an accepted event. This was less true for emergency department diagnosis of ACS or stroke (which were often diagnoses given early on in the clinical presentation before all investigations were completed), and for participant reported heart failure or stroke (which was often contradicted by primary care providers examining their records of the events). However, the largest category of disallowed events was new diagnosis of glaucoma – given ophthalmologists and optometrists appear prone to provide a claims diagnosis of glaucoma at the stage of elevated intra-ocular pressure, prior to any loss of vision. In Figure S3-1 below we show the proportion of each outcome that was rejected by the adjudication committee.

Fig. 3-1

Description: Figure shows the number of reported events accepted and rejected during the adjudication process. Sources of reporting included 1) patient reported (Pt-reported), 2) emergency department diagnosis (ED dx), and 3) most responsible hospital diagnosis (Hospital dx). Methods of reporting are not mutually exclusive (i.e. a single event might be reported by the patient, as well as have an emergency department diagnosis and a most responsible hospital diagnosis).



Fig. 3-1 Cont.



Adjudication Rules Evolved and Used by the Committee

GENERAL RULES

- Each outcome will be individually adjudicated and the committee will be blinded to allocation at all times. All available administrative health data and all available participant reporting of the same outcomes will be simultaneously presented to the adjudication committee for deliberation.
- When the administrative health data and participant self-reported data do not corroborate each other, adjudication will be deferred and the participant's family physician will be contacted and requested to provide details from their medical records as to whether or not the event in question occurred.
- When the family physician is consulted and provides additional data, their response, if definitive, will generally be accepted over all else.
- When additional information from the family physician cannot be collected, and the participant reported that the event happened out of province, where administrative data is not available, the event (if credibly described) will generally be accepted unless stated otherwise.
- If family physicians do not respond to two fax requests for information and one phone call, their supplementary information will be considered not available.
- Each hospitalization will only be attributed to one type of outcome event (e.g. MI/ACS OR HF, but not both).
- Dates from administrative data will be accepted over self-reported dates, when both are available.
- If emergency room and hospitalization dates for the same event type differ substantially, the participant, and then the family physician (if needed) will be contacted to clarify the correct date, and if these are the same event.
- When administrative data for an ED visit, and a hospitalization, appear to indicate the same event, the earlier of the two dates (i.e. the ED visit) will be used.
- If participants do not remember the specific date of an event, but instead report a month, the 15th of that month will be assigned. If they instead report it is "early", or "late" in a typical month then the 7th (early) or 23rd (late) of that month will be assigned as the date of the event.
- One participant was known to have died (reported by family) but the date was unknown. In this case the date of death was assigned to be halfway between the last successful follow-up and the next planned follow-up.
- Only the first occurrence of an accepted adjudicated event will be counted.

DIAGNOSIS SPECIFIC RULES

Acute Coronary Syndrome (ACS) and Heart Failure (HF)

- When a participant has both an ACS diagnosis and a heart failure (HF) diagnosis during the same hospital admission, only the ACS event will be counted.
- When a participant arrives in the ED or hospital with a most responsible diagnosis of angina, whether or not it was diagnosed as unstable angina, it will be considered a potential acute coronary syndrome event that requires adjudication.
- When an ED visit diagnosis was coded as HF, it will be considered a HF event even if the patient or family physician describe other physiologic contributors such as anemia or arrhythmia. The exception to this is a simultaneous ACS diagnosis (in which only ACS will be recorded) or an ultimate alternative diagnosis that could explain the presentation which is not cardiac (e.g. a final diagnosis that the shortness of breath bringing the patient to hospital was caused by a COPD exacerbation, and not HF).
- When participants or family physicians are not available to corroborate *hospitalizations* where the most responsible diagnosis in administrative data is ACS or HF, it will be accepted.
- When participants or family physicians are not available to corroborate *emergency department visits* where the most responsible diagnosis is ACS or HF, and *there is no corresponding hospitalization*, it will be rejected, as many of these are early diagnoses made prior to all investigations being completed.
- When participants or family physicians are not available to elaborate on hospitalizations *without* a corresponding ED visit and the diagnosis is coronary artery bypass grafting, it will not be accepted, as the procedure was potentially treating stable angina.
- When participants or family physicians are not available to elaborate on hospitalizations *with* a corresponding ED visit and the diagnosis is coronary artery bypass grafting, it will be considered an acute coronary syndrome event given the presentation was to the ED.
- When participants dispute an administrative data diagnosis of ACS or HF, and further validation from a family physician is not available, the event will be rejected if it is an ED visit without a corresponding hospitalization. If instead the administrative data diagnosis is a hospitalization with a most responsible diagnosis of ACS or HF, then acceptance will be on a case-by-case basis depending on the other information provided by the participant.
- A diagnosis of MI which is described as being “Type II” in the discharge summary (i.e. demand outstripping supply, rather than an acute arterial obstruction) will be rejected as likely indicating a different clinical entity.

Stroke

- Stroke excludes TIA and requires persistence of neurological symptoms beyond 24 hours, or brain imaging evidence of cerebral infarction or hemorrhage.
- CRAO (Central and branch retinal artery occlusion) is considered a stroke event.

- When participants or family physicians are not available to corroborate hospitalizations where the most responsible diagnosis is stroke, it will be accepted.
- When participants or family physicians are not available to corroborate emergency department visits where the most responsible diagnosis is stroke, and there is no corresponding hospitalization, it will be rejected, as many of these are early diagnoses made prior to all investigations being completed.
- When participants dispute an admin data diagnosis of stroke, and further validation from a family physician is not available, the event will be rejected if it is an ED visit without a corresponding hospitalization. If instead the admin diagnosis is a hospitalization with a most responsible diagnosis of stroke, then acceptance will be on a case-by-case basis depending on the other information provided by the participant.

Unplanned All-cause Hospital and ED Visits

- Because hospitalizations/ED visits are so numerous, we will not attempt to confirm them with family physicians.
- To qualify as “unplanned” we require a diagnosis where a symptom is likely to have led the patient to present themselves to the hospital acutely.
- Where unclear if the event was unplanned, we will err on the side of excluding the event.
- We will disqualify admissions for surgery that are typically elective (e.g. hip or knee replacement for osteoarthritis, cancer surgery) and any ED procedures that are considered elective (e.g. a booked blood transfusion or chemotherapy or “follow-up care”).
- However, if participants present to the emergency department with complications from an elective surgery after discharge, that will be counted as an acute event (eg. Prostate biopsy bleed, bleeding post polypectomy). We will not accept unplanned overnight hospital stays when an elective day surgery was planned (i.e. when the patient did not initially leave the hospital and subsequently present to emergency).
- When cancer is listed as an ED visit diagnosis, it will be accepted. When cancer is listed as the reason for hospitalization but there is no corresponding ED visit, it will be rejected.
- If participants presented to the emergency room but left without being seen, that event will be rejected.
- We excluded one patient presenting to the ED for “medication refill”, and one for the start of “palliative care”.
- If participants describe attending an “urgent care centre” that is not attached to a hospital, it will not be counted as an emergency department, and the event will be excluded.

Hip Fractures

- Any hip fracture, whether or not surgery was involved, will be counted as a fracture event.
- Hip fracture includes fracture of the acetabulum.

- If neither primary care provider information, nor participant interview, are available to corroborate an administrative health indication of hospitalization for hip fracture it will still be accepted.

Death

- If the administrative health data, dialog with friends/family, published obituaries, or primary care providers indicate a death, this data will be accepted unless there is participant follow up data to contradict it.

New Diagnosis of Glaucoma

- It became clear early on that there are many outpatient administrative health diagnoses of glaucoma provided to patients who only have elevated intraocular pressure. The same is true for prescriptions of glaucoma eyedrops, which are often prescribed prior to any of the visual loss that would make the diagnosis of glaucoma clear. As a result, an administrative data diagnosis of glaucoma, or a prescription for glaucoma eye drops, on its own, is insufficient to indicate a new diagnosis of glaucoma.
- If family physicians cannot corroborate or refute an administrative data diagnosis of glaucoma using ophthalmology or optometry consult letters, we require participants to have been told by their ophthalmologist or optometrist that they have glaucoma in order to accept the diagnosis.
- If participants were told they have glaucoma by an ophthalmologist or optometrist and there are no family physician records to review, the diagnosis will be accepted. If the family physician did have an ophthalmology or optometry report to review and glaucoma is not mentioned, the diagnosis will be rejected.
- “Glaucoma suspect” in an ophthalmology or optometry report will be accepted as a glaucoma diagnosis.

Nursing Home admission

- Both administrative health data and participant/family-reports indicating nursing home admission will be accepted. Corroboration from family physicians is not required.
- Participants who began the trial in a low level of care assisted living facility where they had control of their own medication administration time, who subsequently move to a higher level of care where medications are administered by nurses or care aides, will be accepted as being admitted to a nursing home.

Non-vertebral fractures

- Includes any fracture, other than vertebral (back or neck), that is presumed to stem from trauma.
- We will include avulsion fractures as being traumatic, and we will exclude stress fractures.

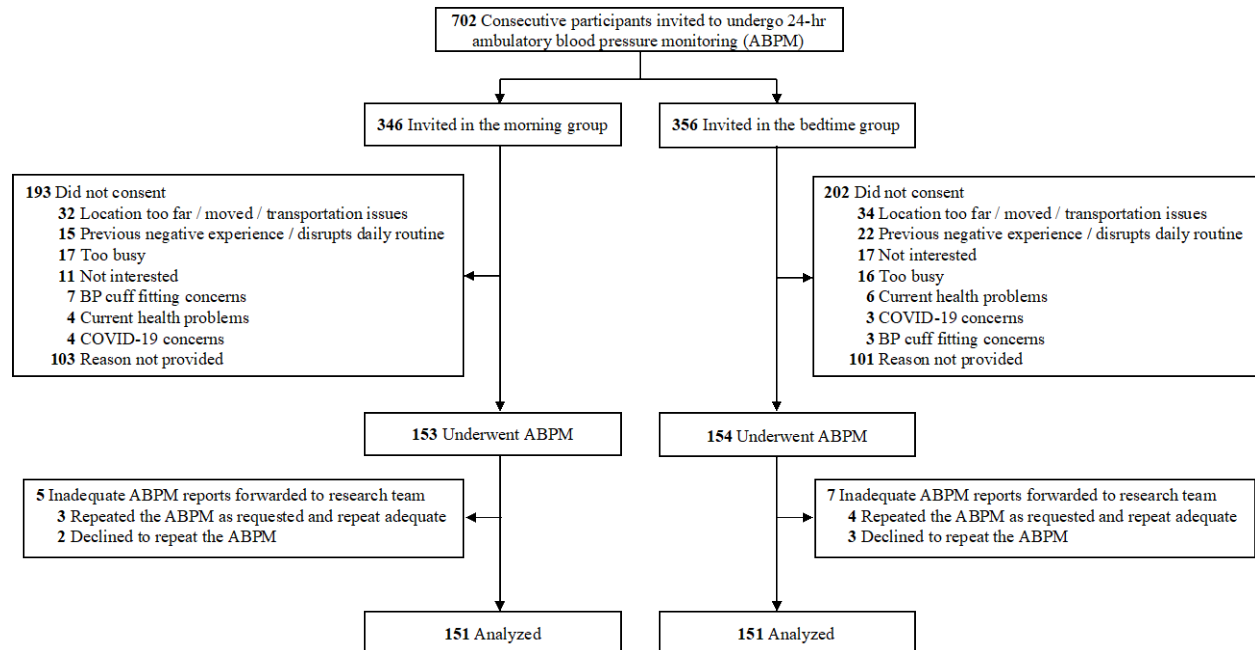
- Fractures that are attributed to osteomyelitis, or malignancy, will be excluded.
- Participants need to receive their fracture diagnosis from a health care practitioner (i.e. self-diagnosis will not be accepted).
- Includes sacral / pelvic fractures
- If information describing a fracture is ultimately only available from one source (i.e. patient report or administrative health data) it will generally be accepted.

SECTION 4: Ambulatory Blood Pressure Monitoring

Although we did not measure BP in all participants, a consecutive sample of 151 bedtime, and 151 morning group volunteers residing in one of six Alberta communities (Calgary, Edmonton, Camrose, Sherwood Park, Spruce Grove, and Red Deer) underwent 24-hour ambulatory BP monitoring (ABPM) carried out by a local cardiology diagnostics provider (Advanced Respiratory Care Network - www.arcnetwork.ca/cardiology-diagnostics). The intention was to carry this assessment out 6-months post-randomization, however a variety of logistic hurdles, including the start of the COVID-19 pandemic, substantially delayed these assessments for many individuals – instead taking place a median 9.6 months (IQR 7.1 to 29.2 months) into the trial.

A Suntech Oscar 2 M250 ambulatory BP monitor was employed in all cases, with settings per the local provider's usual routine. Although the frequency of BP readings varied somewhat by location, this typically entailed taking BP readings every 30 minutes during the day and every 45 minutes overnight, with the anticipated sleep time window pre-defined by the participant (most often 11:00 PM to 7:00 AM). When ABPM output was grossly inadequate, the cardiology diagnostics provider performing the service asked participants to repeat those assessments per their usual routine, without interacting with the research team. ABPM results forwarded to the research team, who required a minimum of 5 successful overnight readings and 8 successful daytime readings, were additionally considered inadequate for interpretation for seven bedtime and five morning participants, each of whom was asked to repeat the assessment. Of these, 3/7 bedtime and 2/5 morning participants declined a repeat evaluation, which required the same number of additional participants (consecutively selected) to be invited to undergo 24-hour monitoring to replace them. Overall, in order to obtain 151 consecutive participants from each arm of the trial, we needed to invite 356 bedtime group participants and 346 morning group participants (Figure 4-1 below).

Fig. 4-1 Flow diagram for 24-hour ambulatory blood pressure monitoring



SECTION 5: Definitions, Statistical Analyses, and Changes to Secondary Outcomes

Lost to follow-up: Participants were considered lost to follow-up when attempts to re-establish contact were exhausted. This included, but was not limited to, initially reaching out to all provided alternate contacts, and then contacting their primary care provider's clinic.

Three situations initiated the process of reaching out to alternate contacts:

1. The study team attempted to contact the participant by phone or email 10 times without a reply.
2. The participant missed 2 interviews in a row (“missed” being defined as having still been unreachable 5 weeks after the scheduled interview date).
3. The study team phoned the participant, but the phone line was disconnected or non-functional.

If alternate contacts were not reached within a week, the participant’s primary care clinic was contacted by phone to inquire about the participant’s status and contact information. On rare occasions when the primary care office could not be contacted (i.e. the provider had retired, the clinic had closed, or the participant had changed primary care providers without notifying us) the study team mailed a letter to the participant's current address in an effort to re-establish contact.

Analytic Covariates: Analytic covariates were predefined and unique to each outcome. The number of covariates used, and the order in which they were selected, was provided in the published statistical analysis plan for every outcome examined. For the primary analysis, baseline characteristics used as covariates to produce the adjusted hazard ratio included: Age, sex, physical frailty score (Tilburg), current smoker, number of non-BP medications, Overall Health Score (EQ-5D-5L), prior 6 months hospitalization, heart failure, diabetes, coronary artery disease, stroke or TIA, chronic kidney disease, dialysis, BMI >35kg/m², BMI <20kg/m², sleep apnea, number of exercise days, and province of residence.

Appropriateness of the statistical analyses used: Categorical variables were summarized by count and percent, while distributions of recorded variables were described by appropriate measures of center (mean, median) and spread (SD, IQR). Methods of analyses to compare between groups were applied according to a published trial protocol and analysis plan.

Data were considered missing at random. Covariate data were replaced by the modal observation for categorical variables and by the mean for continuous ones. No outcome data were imputed.

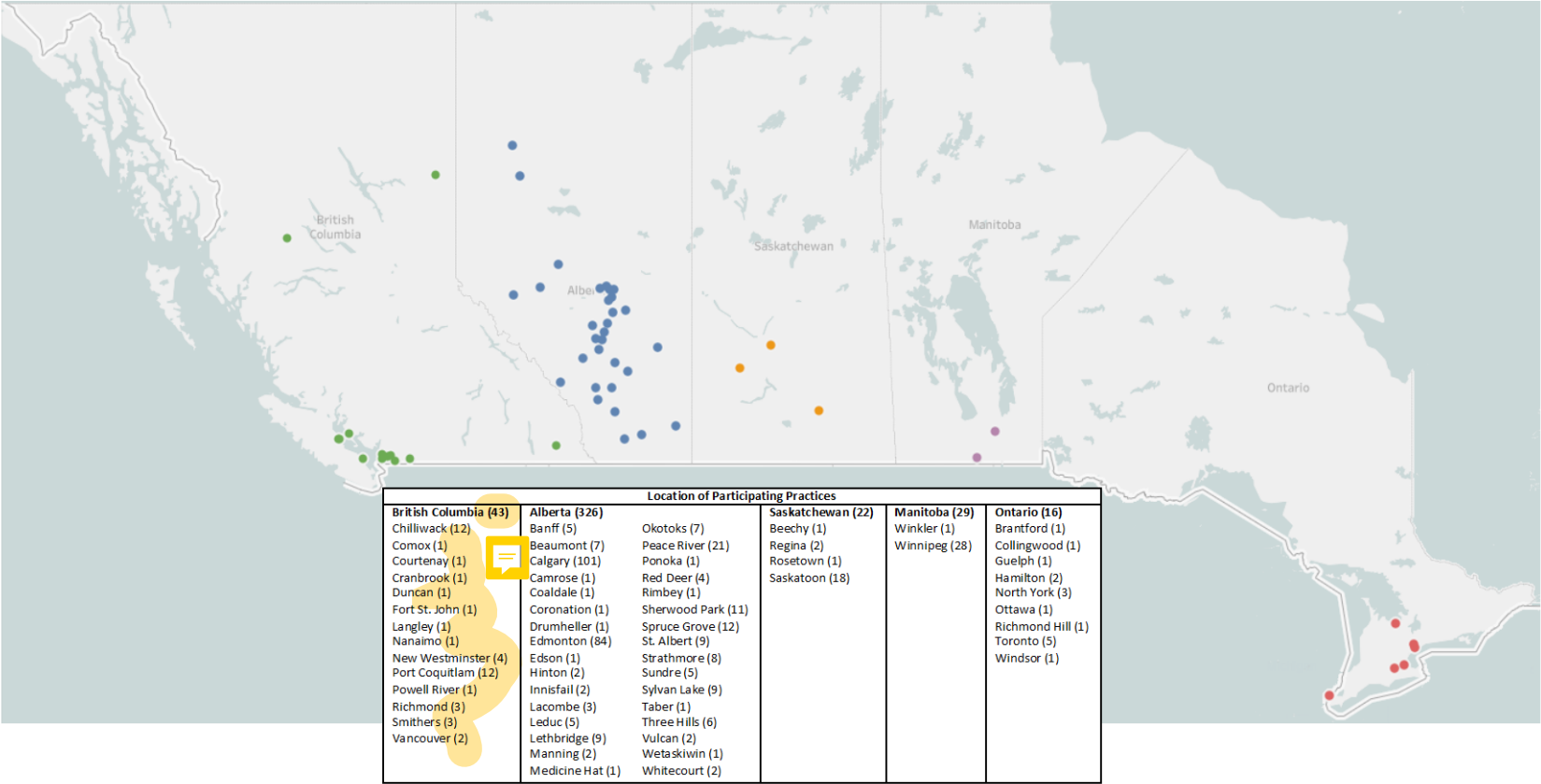
A series of Cox Proportional hazards models were used to assess the primary and secondary outcomes over and above the effects of clinical and demographic factors. Where group sizes / number of outcomes were limited, reduced models were used to avoid overfitting. Variables were included in a predefined order of presumed importance. We confirmed no violations of the proportional hazards assumption in the allocation group using a Kolmogorov-type supremum test on the sum of martingale residuals on 1000 simulated patterns for all outcomes. Any evidence of

proportional hazard assumption violation among covariates was considered insignificant because they are not of primary interest, and adjustment models did not appreciably change the results compared to unadjusted analysis.

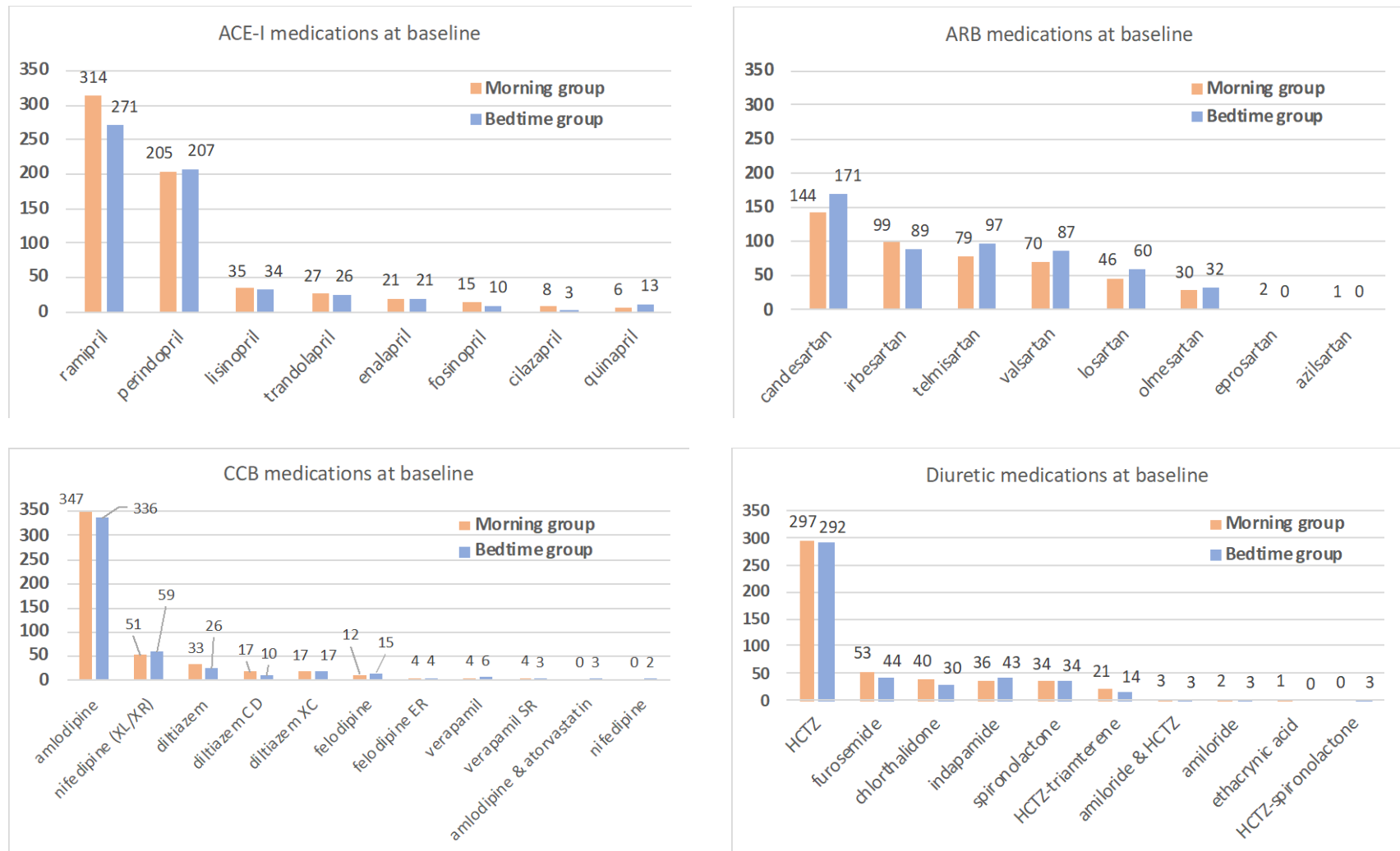
Changes to secondary outcomes: The following changes were made to secondary outcomes while fully blinded to all primary and secondary outcomes. They came about at the time the draft manuscript for the published protocol was being written and co-investigators were reviewing and providing feedback on the current version of the protocol.

1. **Change from baseline in self-reported Overall Health Score at 12-months was altered to Overall Health Score at 12-months.** This alteration was made when it was newly recognized that comparing absolute change (or percentage change) from baseline between groups often provides less power to detect a difference than comparing on-treatment scores. (Vickers AJ. The use of percentage change from baseline as an outcome in a controlled trial is statistically inefficient: a simulation study. BMC Medical Research Methodology 2001;1:6).
2. **Total cost of care was added as an outcome** to strengthen the cost analysis –whereas previously only acute care costs were to be analyzed.
3. **Non-vertebral fracture was added as a secondary outcome.** Although captured by administrative health data throughout the trial, we had not been asking participants about fractures other than hip fracture until mid-trial when we began to ask about any fractures. Hence this outcome was less sensitively captured in the early years of the trial.
4. **Syncope in the last month, falling in the last month, new glaucoma diagnosis, nocturia frequency, and new impairment consistent with dementia were added as secondary safety outcomes.** The information defining these outcomes had been collected from all participants since the beginning of the trial but had not previously been prespecified as outcomes we would analyze. They were all believed to meaningfully augment our other safety outcomes.

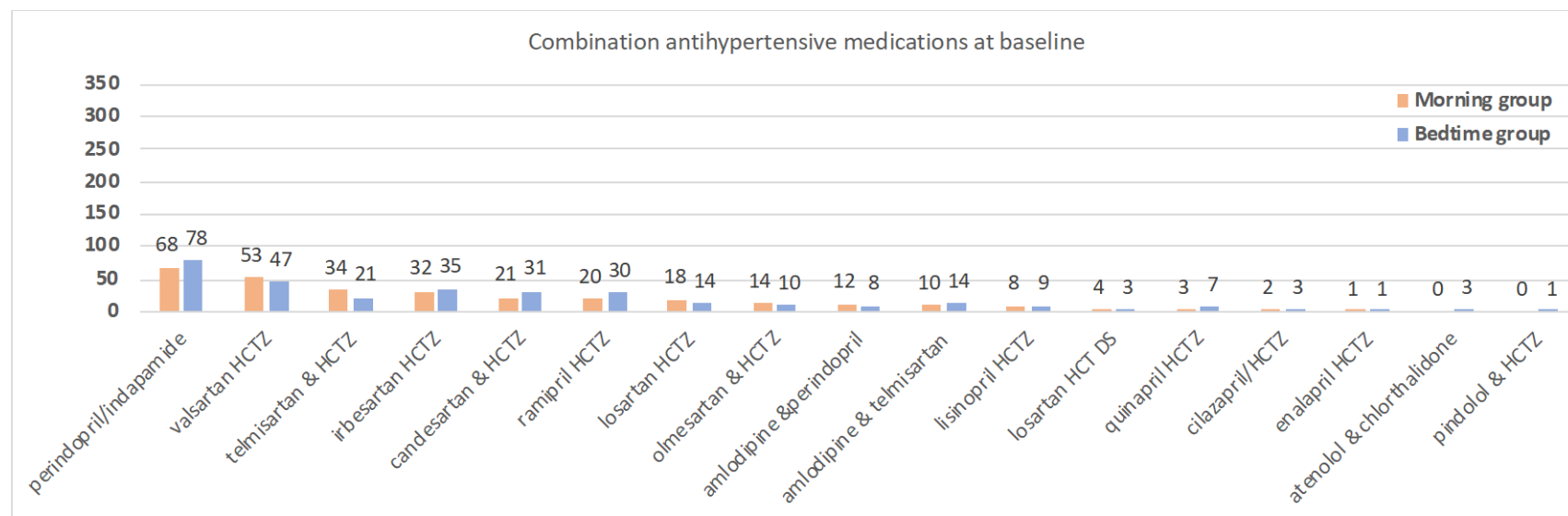
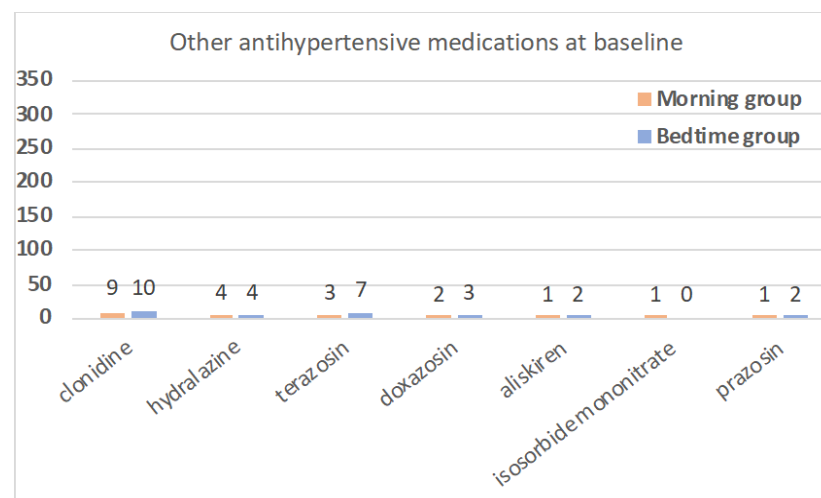
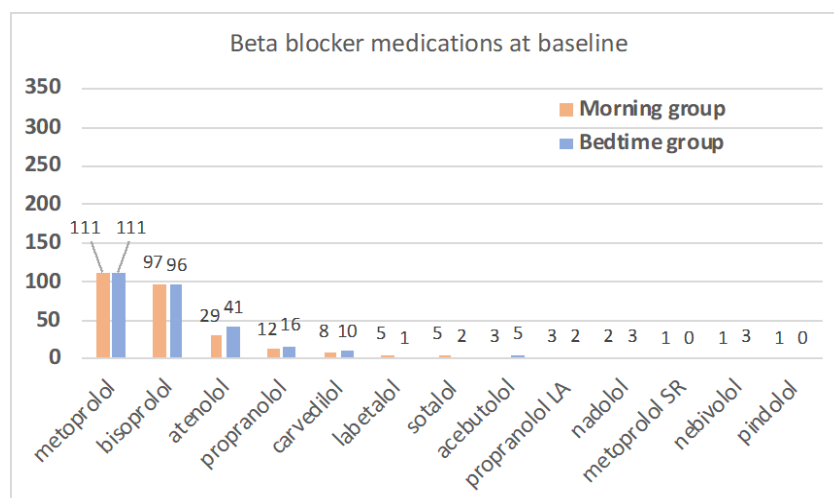
eFigure 1. Location of participating medical clinics



eFigure 2. Medications in use at baseline

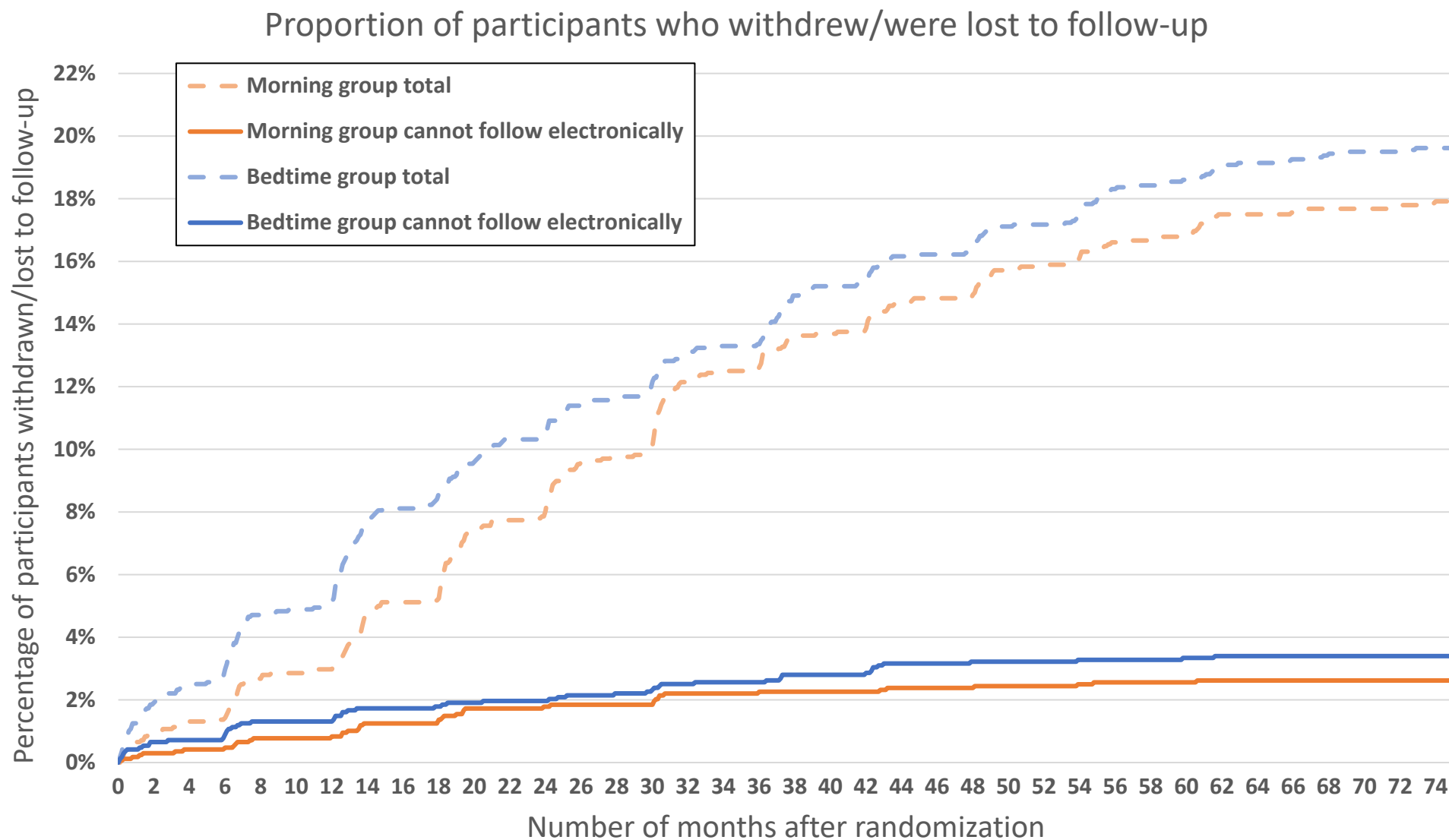


eFigure 2. Medications in use at baseline (continued)



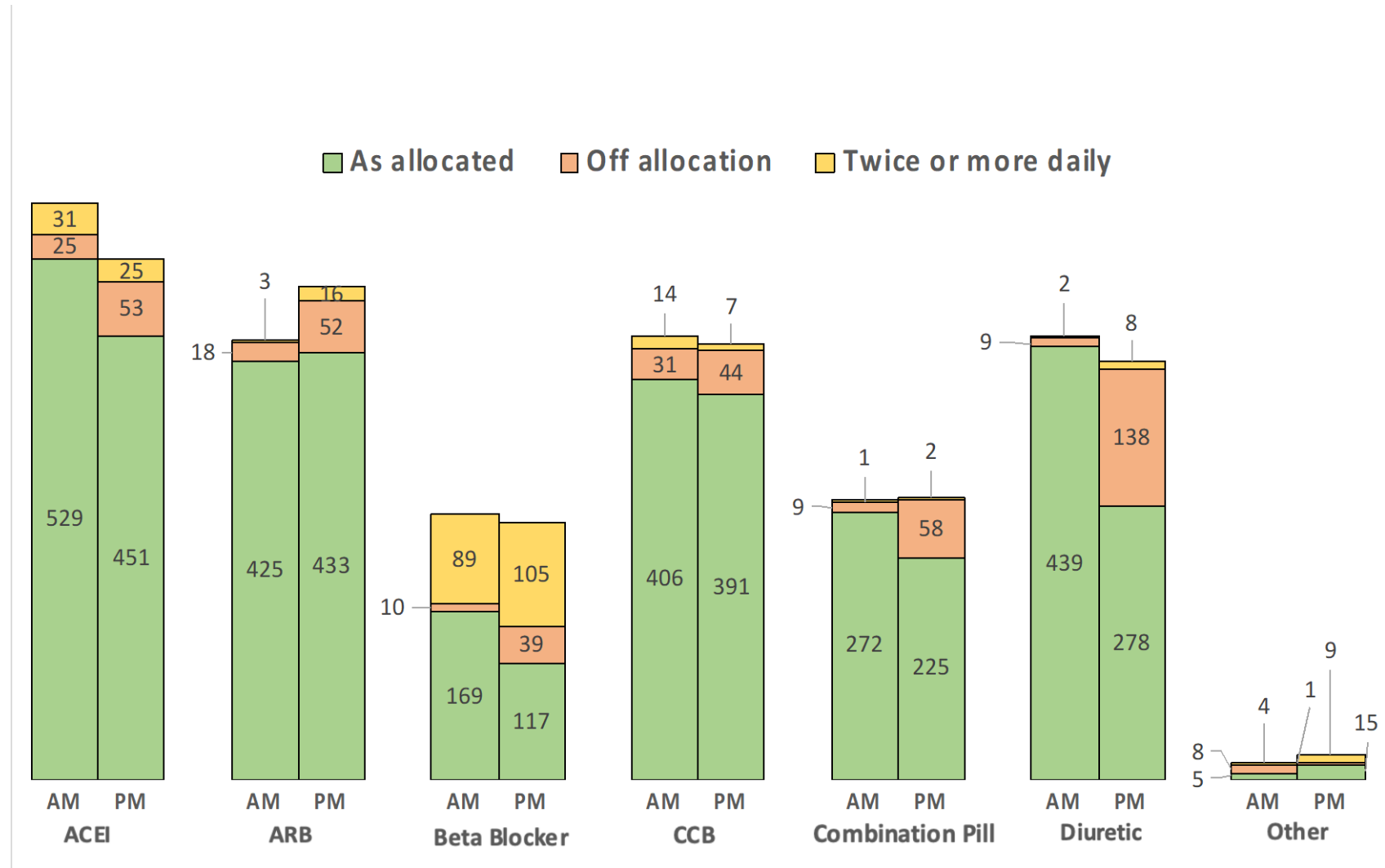
eFigure 3. Proportion of participants who withdrew or were lost to follow-up

Figure shows the proportion of participants from each allocation group that withdrew or were lost to follow-up over the duration of the trial. Dashed lines are all participants withdrawing or lost to follow-up – many of whom we could continue to follow using administrative health data. Solid lines are participants withdrawing or lost to follow-up who we could not continue to follow. Only 3.4% of bedtime group participants and 2.6% of morning group participants could not be followed until the end of the trial.



eFigure 4. Adherence to allocation time at 6-months, by medication class

Figure shows self-reported adherence to allocated administration time at 6-months for all medications used by those in the bedtime group (indicated by PM on the x-axis), and all those in the morning group (indicated by AM on the x-axis)



eTable 1. Study recruitment methods

Recruitment method	Morning (N=1680)	Bedtime (N=1677)
Primary care provider (PCP)*	1295 (77.1%)	1273 (75.9%)
Social media	249 (14.8%)	256 (15.3%)
News media	89 (5.3%)	96 (5.7%)
Other†	47 (2.8%)	52 (3.1%)
* PCP recruitment included mailing a letter of introduction / study information to all attached patients with hypertension, clinic posters, and tear away notepads describing the trial that could be handed to patients during appointments. 436 PCPs mailed recruitment information (429 family physicians and 7 nurse practitioners). 41,128 letters were mailed in total. A median of 77.5 letters were mailed per PCP (IQR 77.25). Overall, 6.2% of all letters mailed resulted in a randomized participant. † Other includes: 1) word of mouth, 2) a website describing the trial (i.e. the BedMed study website, the pragmatic trials collaborative website, or the University of Alberta website), 3) public events, or 4) the same social media ad instead running on closed circuit television within Alberta government run outpatient clinics.		

eTable 2. Proportion of interviews completed online vs over the phone

	Week 1*	Week 6	Month 6	Month 12	Month 18	Month 18 (SBT) [†]	Month 24	Month 30	Month 36	Month 42	Month 48	Month 54	Month 60	Month 66	Month 72
Online no. (%)	N/A	N/A	1982 (62.9)	1897 (60.9)	1727 (57.0)	N/A	1638 (58.1)	1445 (57.5)	1276 (57.8)	1217 (58.7)	986 (56.8)	806 (53.1)	568 (52.4)	300 (46.8)	183 (49.2)
Phone no. (%)	2038 (100)	3275 (100)	1171 (37.1)	1219 (39.1)	1304 (43.0)	2941 (100)	1180 (41.9)	1068 (42.5)	931 (42.2)	856 (41.3)	751 (43.2)	712 (46.9)	517 (47.6)	341 (53.2)	189 (50.8)
Total number of interviews	2038	3275	3153	3116	3031	2941	2818	2513	2207	2073	1737	1518	1085	641	372
*Week 1 interview was only required if the participant needed to make a medication timing change. Otherwise, no Week 1 interview occurred. †Month-18 Short Blessed Test (SBT) was either completed along with the Month-18 interview or separately for those that completed the Month-18 interview online.															

eTable 3. Expanded baseline characteristics

Characteristic	Treatment (Bedtime) N = 1677	Control (Morning) N = 1680	Overall N = 3357
Age – yr, median (IQR)	67 (60, 73)	67 (61, 73)	67 (60,73)
<50 yr – no. (%)	73 (4.4)	75 (4.5)	148 (4.4)
50-59 yr – no. (%)	308 (18.4)	293 (17.4)	601 (17.9)
60-69 yr – no. (%)	624 (37.2)	630 (37.5)	1254 (37.4)
70-79 yr – no. (%)	519 (30.9)	532 (31.7)	1051 (31.3)
≥ 80 yr – no. (%)	153 (9.1)	150 (8.9)	303 (9.0)
Male – no. (%)	727 (43.4)	737 (43.9)	1464 (43.6)
Education – no. (%)			
University degree	693 (41.3)	690 (41.1)	1383 (41.2)
Technical/Trade college diploma	437 (26.1)	413 (24.6)	850 (25.3)
High School	454 (27.1)	454 (27)	908 (27.0)
Less than high school	91 (5.4)	122 (7.3)	213 (6.3)
Declined to answer or missing	2 (0.1)	1 (0.1)	3 (0.1)
Ethnicity – no. (%)			
White	1565 (93.3)	1587 (94.5)	3152 (93.9)
Asian	42 (2.5)	34 (2.0)	76 (2.3)
Indigenous	29 (1.7)	22 (1.3)	51 (1.5)
South East Asian/Indian	17 (1.0)	20 (1.2)	37 (1.1)
Black/African American	7 (0.4)	7 (0.4)	14 (0.4)
Latino	5 (0.3)	4 (0.2)	9 (0.3)
Other	9 (0.5)	5 (0.3)	14 (0.4)
Declined to answer	3 (0.2)	1 (0.1)	4 (0.1)
Province – no. (%)			
Alberta	1200 (71.5)	1200 (71.4)	2400 (71.5)
British Columbia	235 (14.0)	236 (14.0)	471 (14.0)
Manitoba	130 (7.8)	131 (7.8)	261 (7.8)
Ontario	74 (4.4)	75 (4.5)	149 (4.4)
Saskatchewan	38 (2.3)	38 (2.3)	76 (2.2)
Chronotype – no. (%)			
Early Bird	881 (52.5)	859 (51.1)	1740 (51.8)
Night Owl	477 (28.5)	495 (29.5)	972 (29.0)
Neither	319 (19.0)	326 (19.4)	645 (19.2)
Currently smokes – no. (%)	122 (7.3)	121 (7.2)	243 (7.2)
# exercise days / week, median (IQR) [†]	3 (0, 5)	3 (0, 5)	3 (0, 5)
No exercise – no. (%) [†]	434 (25.9)	458 (27.3)	892 (26.6)

Characteristic	Treatment (Bedtime) N = 1677	Control (Morning) N = 1680	Overall N = 3357
BMI, median (IQR)	28.8 (25.7, 33.0)	28.9 (25.7, 32.9)	28.8 (25.7, 32.9)
BMI group – no. (%)			
BMI < 20	21 (1.3)	20 (1.2)	41 (1.2)
BMI > 35	290 (17.3)	284 (16.9)	574 (17.1)
Physically frail – no. (%) [‡]	299 (17.8)	307 (18.3)	606 (18.1)
Overall Health Score, median (IQR) [§]	80 (70, 90)	80 (75, 90)	80 (73, 90)
Overall Health Score group – no. (%)			
<50	61 (3.6)	54 (3.2)	115 (3.4)
50-59	91 (5.4)	98 (5.8)	189 (5.6)
60-69	139 (8.3)	139 (8.3)	278 (8.3)
70-79	413 (24.6)	373 (22.2)	786 (23.4)
80-89	517 (30.8)	510 (30.4)	1027 (30.6)
≥90	456 (27.2)	506 (30.1)	962 (28.7)
Short Blessed Test score – no. (%)			
≤3	1376 (82.1)	1403 (83.5)	2779 (82.8)
4-6	248 (14.8)	224 (13.3)	472 (14.1)
7-9	33 (2)	29 (1.7)	62 (1.8)
≥10	19 (1.1)	23 (1.4)	42 (1.3)
Social deprivation index quintiles – no. (%) [¶]			
1 (most privileged)	566 (33.8)	572 (34)	1138 (33.9)
2	405 (24.2)	383 (22.8)	788 (23.5)
3	308 (18.4)	296 (17.6)	604 (18.0)
4	193 (11.5)	204 (12.1)	397 (11.8)
5 (least privileged)	73 (4.3)	91 (5.4)	164 (4.9)
Unknown	132 (7.9)	134 (8)	266 (7.9)
Rural – no. (%) ^{**}	201 (12)	219 (13)	420 (12.5)
Comorbidities – no. (%)			
Sleep apnea	377 (22.5)	341 (20.3)	718 (21.4)
Diabetes	289 (17.2)	311 (18.5)	600 (17.9)
CAD	172 (10.3)	188 (11.2)	360 (10.7)
CKD	119 (7.1)	129 (7.7)	248 (7.4)
COPD	86 (5.1)	80 (4.8)	166 (4.9)
Stroke	75 (4.5)	75 (4.5)	150 (4.5)
TIA	70 (4.2)	54 (3.2)	124 (3.7)
Dementia ^{††}	46 (3.0)	48 (3.1)	94 (3.1)
Heart Failure	28 (1.7)	32 (1.9)	60 (1.8)
Hip fracture	22 (1.3)	27 (1.6)	49 (1.5)
Dialysis	3 (0.2)	2 (0.1)	5 (0.1)

Characteristic	Treatment (Bedtime) N = 1677	Control (Morning) N = 1680	Overall N = 3357
Hospitalization in prior 6-months – no. (%)	95 (5.7)	100 (6)	195 (5.8)
Number of medications – no. (%)			
Blood Pressure - lowering			
1	895 (53.4)	908 (54.0)	1803 (53.7)
2	588 (35.1)	577 (34.3)	1165 (34.7)
3	155 (9.2)	170 (10.1)	325 (9.7)
≥4	39 (2.3)	25 (1.5)	64 (1.9)
Total daily prescription meds			
1	227 (13.5)	225 (13.4)	452 (13.5)
2	346 (20.6)	341 (20.3)	687 (20.5)
3	296 (17.7)	312 (18.6)	608 (18.1)
4	274 (16.3)	259 (15.4)	533 (15.9)
5	199 (11.9)	201 (12)	400 (11.9)
≥6	335 (20)	342 (20.4)	677 (20.2)
Total daily non-prescription meds			
0	1028 (61.3)	1011 (60.2)	2039 (60.1)
1	554 (33)	570 (33.9)	1124 (33.5)
2	80 (4.8)	82 (4.9)	162 (4.8)
3	11 (0.7)	13 (0.8)	24 (0.7)
≥4	4 (0.2)	4 (0.2)	8 (0.2)
Vitamins/herbal supplements			
0	477 (28.4)	423 (25.2)	900 (26.8)
1	330 (19.7)	369 (22)	699 (20.8)
2	310 (18.5)	312 (18.6)	622 (18.5)
3	233 (13.9)	227 (13.5)	460 (13.7)
≥4	327 (19.5)	349 (20.8)	676 (20.1)

Characteristic	Treatment (Bedtime) N = 1677	Control (Morning) N = 1680	Overall N = 3357
Type of BP medications – no. (%)			
ACE inhibitor	584 (34.8)	631 (37.6)	1215 (36.2)
Angiotensin II receptor blocker	536 (32)	471 (28)	1007 (30.0)
Calcium channel blocker	479 (28.2)	489 (29.1)	968 (28.8)
Diuretic	446 (26.6)	472 (28.1)	918 (27.3)
Combination pill	315 (18.8)	300 (17.9)	615 (18.3)
Beta-blocker	289 (17.2)	278 (16.5)	567 (16.9)
Other class of BP med	26 (1.6)	21 (1.3)	47 (1.4)
* Percentages may not total 100 if data was missing. IQR denotes interquartile range. † “How many days in the past week have you exercised for 30 min or more, vigorously enough to raise your breathing rate?” Participant considered sedentary if answer = 0. ‡ As determined by the physical frailty subscale of the Tilburg questionnaire. Subscale score ≥ 3 = physically frail. § Self-reported quality of life via EQ5D5L. Higher numbers indicate higher quality of life. Range 0 – 100. A weighted, six-item test of cognitive impairment. Scores range from 0 – 28. A score of 5 – 9 suggests questionable impairment and scores of ≥ 10 suggest impairment consistent with dementia. ¶ Postal code derived social deprivation index (Pampalon) ** Postal code derived (as per Canada Post designation) †† Derived exclusively from administrative health data – hence unavailable for participants from Saskatchewan and Ontario (% calculated using only those for whom there was data in the denominator)			

eTable 4. Baseline characteristics of those completing the trial versus those withdrawing or lost to follow-up

Characteristic	Completed study or death (n=2726)	Withdrew or lost to follow up (n=631)	p-value
Median age (IQR) – yr	67 (61 – 73)	68 (60 – 75)	0.32
Age group – no. (%)			<0.001
<50 yr	110 (4)	38 (6)	
50-59 yr	484 (17.8)	117 (18.5)	
60-69 yr	1048 (38.4)	206 (32.6)	
70-79 yr	868 (31.8)	183 (29)	
≥80 yr	216 (7.9)	87 (13.8)	
Sex – no. (%)			0.11
Male	1207 (44.3)	257 (40.7)	
Female	1519 (55.7)	374 (59.3)	
Chronotype – no. (%)			0.09
Early Bird	1388 (50.9)	352 (55.8)	
Night Owl	818 (30)	154 (24.4)	
Neither	520 (19.1)	125 (19.8)	
Education – no. (%)			<0.001
University degree	1174 (43.1)	209 (33.1)	
High School	714 (26.2)	194 (30.7)	
Technical/Trade college diploma	692 (25.4)	158 (25)	
Less than high school	145 (5.3)	68 (10.8)	
Declined to answer	1 (<0.1)	2 (0.3)	
Ethnicity – no. (%)			0.008
White/Caucasian	2564 (94.1)	588 (93.2)	
Asian	62 (2.3)	14 (2.2)	
Indigenous	42 (1.5)	9 (1.4)	
South East Asian/Indian	29 (1.1)	8 (1.3)	
Black/African American	10 (0.4)	4 (0.6)	
Latino	8 (0.3)	1 (0.2)	
Other	11 (0.4)	3 (0.5)	
Declined to answer	0 (0)	4 (0.6)	
Province – no. (%)			<0.001
Alberta	1900 (69.7)	500 (79.2)	
British Columbia	406 (14.9)	65 (10.3)	
Saskatchewan	67 (2.5)	9 (1.4)	
Manitoba	221 (8.1)	40 (6.3)	
Ontario	132 (4.8)	17 (2.7)	
Currently smokes – no. (%)	180 (6.6)	63 (10)	0.003

Characteristic	Completed study or death (n=2726)	Withdrew or lost to follow up (n=631)	p-value
Median # exercise days / week (IQR) [†]	3 (0 – 5)	3 (0 – 5)	0.07
No exercise – no. (%) [†]	706 (25.9)	186 (29.5)	0.07
Median BMI (IQR)	28.7 (25.7 – 32.9)	29.1 (25.8 – 33.1)	0.42
BMI group – no. (%)			
BMI < 20	29 (1.1)	12 (1.9)	0.08
BMI > 35	471 (17.3)	103 (16.3)	0.57
Physically frail – no. (%) [‡]	458 (16.8)	148 (23.5)	<0.001
Median overall health score (IQR) [§]	80 (75 – 90)	80 (70 – 90)	<0.001
Overall health score group – no. (%)			<0.001
<50	77 (2.8)	38 (6)	
50-59	143 (5.2)	46 (7.3)	
60-69	217 (8)	61 (9.7)	
70-79	637 (23.4)	149 (23.6)	
80-89	860 (31.5)	167 (26.5)	
≥90	792 (29.1)	170 (26.9)	
SBT score group – no. (%)			<0.001
≤3	2290 (84)	489 (77.5)	
4-6	364 (13.4)	108 (17.1)	
7-9	45 (1.7)	17 (2.7)	
≥10	25 (0.9)	17 (2.7)	
Social deprivation index quintiles – no. (%) [¶]			0.31
1 (most privileged)	931 (34.2)	207 (32.8)	
2	651 (23.9)	137 (21.7)	
3	484 (17.8)	120 (19)	
4	323 (11.8)	74 (11.7)	
5 (least privileged)	134 (4.9)	30 (4.8)	
Unknown	203 (7.4)	63 (10)	
Rural – no. (%) ^{**}	340 (12.5)	80 (12.7)	0.89
Prior 6-month hospitalization – no. (%)	153 (5.6)	42 (6.7)	0.31
Key comorbidities – no. (%)			
Sleep apnea	588 (21.6)	130 (20.6)	0.59
Diabetes	488 (17.9)	112 (17.7)	0.93
CAD	288 (10.6)	72 (11.4)	0.54
CKD	199 (7.3)	49 (7.8)	0.69
COPD	134 (4.9)	32 (5.1)	0.87
Stroke	117 (4.3)	33 (5.2)	0.30
Dementia ^{††}	67 (2.7)	27 (4.5)	0.02

Characteristic	Completed study or death (n=2726)	Withdrew or lost to follow up (n=631)	p-value
Heart Failure	41 (1.5)	19 (3)	0.01
Hip fracture	42 (1.5)	7 (1.1)	0.42
Dialysis	4 (0.1)	1 (0.2)	0.95
Number of medications – no. (%)			
Blood pressure-lowering			0.15
1	1462 (53.6)	341 (54)	
2	963 (35.3)	202 (32)	
3	251 (9.2)	74 (11.7)	
≥4	50 (1.8)	14 (2.2)	
Non-blood pressure-lowering			0.21
0	539 (19.8)	105 (16.6)	
1	691 (25.3)	169 (26.8)	
2	543 (19.9)	115 (18.2)	
3	348 (12.8)	94 (14.9)	
≥4	605 (22.2)	148 (23.5)	
Total daily prescription medications			0.05
1	383 (14)	69 (10.9)	
2	542 (19.9)	145 (23)	
3	506 (18.6)	102 (16.2)	
4	433 (15.9)	100 (15.8)	
5	330 (12.1)	70 (11.1)	
≥6	532 (19.5)	145 (23)	
Over the counter non-prescription			0.19
0	1658 (60.8)	381 (60.4)	
1	917 (33.6)	207 (32.8)	
2	127 (4.7)	35 (5.5)	
3	20 (0.7)	4 (0.6)	
≥4	4 (0.1)	4 (0.6)	
Vitamins/herbal supplements			0.01
0	738 (27.1)	162 (25.7)	
1	539 (19.8)	160 (25.4)	
2	526 (19.3)	96 (15.2)	
3	375 (13.8)	85 (13.5)	
≥4	548 (20.1)	128 (20.3)	
Type of blood pressure-lowering medication – no. (%)			
ACE inhibitor	1011 (37.1)	204 (32.3)	0.03
Angiotensin II receptor blocker	815 (29.9)	192 (30.4)	0.79
Calcium channel blocker	774 (28.4)	194 (30.7)	0.24

Characteristic	Completed study or death (n=2726)	Withdrew or lost to follow up (n=631)	p-value
Diuretic	741 (27.2)	177 (28.1)	0.66
Combination BP med	488 (17.9)	127 (20.1)	0.19
Beta-blocker	456 (16.7)	111 (17.6)	0.60
Other class of BP med	38 (1.4)	9 (1.4)	0.95
Polypharmacy – no. (%)‡‡	862 (31.6)	215 (34.1)	0.23
Resistant hypertension – no. (%)§§	301 (11)	88 (13.9)	0.04

* Percentages may not total 100 if data was missing. IQR denotes interquartile range.
 † “How many days in the past week have you exercised for 30 min or more, vigorously enough to raise your breathing rate?” Participant considered sedentary if answer = 0.
 ‡ As determined by the physical frailty subscale of the Tilburg questionnaire. Subscale score ≥ 3 = physically frail.
 § Self-reported quality of life via EQ5D5L. Higher numbers indicate higher quality of life. Range 0 – 100.
 || A weighted, six-item test of cognitive impairment. Scores range from 0 – 28. A score of 5 – 9 suggests questionable impairment and scores of ≥ 10 suggest impairment consistent with dementia.
 ¶ Postal code derived social deprivation index (Pampalon)
 ** Postal code derived (as per Canada Post designation)
 †† Derived exclusively from administrative health data – hence unavailable for participants from Saskatchewan and Ontario (% calculated using only those for whom there was data)
 ‡‡ Taking ≥ 5 daily prescription medications.
 §§ Taking ≥ 3 daily BP-lowering medications.

eTable 5. Baseline characteristics for those who did not complete follow-up and could not be followed using administrative data, stratified by study allocation

Characteristic	Morning Allocation (n=44)	Bedtime Allocation (n=57)	p-value
Median age (IQR) – yr	70 (60, 78)	72 (66, 78)	0.37
Age group – no. (%)			0.59
<50 yr	2 (4.5)	1 (1.8)	
50-59 yr	9 (20.5)	6 (10.5)	
60-69 yr	11 (25.0)	16 (28.1)	
70-79 yr	16 (36.4)	24 (42.1)	
≥80 yr	6 (13.6)	10 (17.5)	
Sex – no. (%)			0.52
Male	19 (43.2)	21 (36.8)	
Female	25 (56.8)	36 (63.2)	
Chronotype – no. (%)			0.03
Early Bird	24 (54.5)	28 (49.1)	
Night Owl	17 (38.6)	14 (24.6)	
Neither	3 (6.8)	15 (26.3)	
Education – no. (%)			0.13
University degree	19 (43.2)	15 (26.3)	
High School	10 (22.7)	23 (40.4)	
Technical/Trade college diploma	10 (22.7)	14 (24.6)	
Less than high school	5 (11.4)	3 (5.3)	
Declined to answer	0 (0.0)	2 (3.5)	
Ethnicity – no. (%)			0.82
White/Caucasian	40 (90.9)	53 (93.0)	
Asian	1 (2.3)	1 (1.8)	
Indigenous	1 (2.3)	1 (1.8)	
South East Asian/Indian	1 (2.3)	1 (1.8)	
Black/African American	0 (0.0)	1 (1.8)	
Latino	1 (2.3)	0 (0.0)	
Other	40 (90.9)	53 (93.0)	
Declined to answer	1 (2.3)	1 (1.8)	
Province – no. (%)			0.96
Alberta	25 (56.8)	33 (57.9)	
British Columbia	4 (9.1)	6 (10.5)	
Saskatchewan	5 (11.4)	4 (7.0)	
Manitoba	3 (6.8)	4 (7.0)	
Ontario	7 (15.9)	10 (17.5)	
Currently smokes – no. (%)	2 (4.5)	5 (8.8)	0.41
Median # exercise days / week (IQR) [†]	2 (1, 3)	3 (1, 4)	0.54

Sedentary – no. (%) [†]	10 (22.7)	14 (24.6)	0.83
Median BMI (IQR)	27.6 (25.4, 31.4)	29.1 (25.8, 33.8)	0.22
BMI group – no. (%)			
BMI < 20	1 (2.3)	3 (5.3)	0.45
BMI > 35	4 (9.1)	7 (12.3)	0.61
Physically frail – no. (%) [‡]	12 (27.3)	15 (26.3)	0.91
Median overall health score (IQR) [§]	80 (67, 90)	80 (70, 90)	0.74
Overall health score group – no. (%)			0.28
<50	4 (9.1)	0 (0.0)	
50-59	3 (6.8)	7 (12.3)	
60-69	4 (9.1)	7 (12.3)	
70-79	10 (22.7)	14 (24.6)	
80-89	8 (18.2)	11 (19.3)	
≥90	15 (34.1)	18 (31.6)	
SBT score group – no. (%)			0.27
≤3	36 (81.8)	42 (73.7)	
4-6	8 (18.2)	12 (21.1)	
7-9	0 (0.0)	0 (0.0)	
≥10	0 (0.0)	3 (5.3)	
Social deprivation index quintiles – no. (%) [¶]			0.52
1 (most privileged)	16 (36.4)	21 (36.8)	
2	7 (15.9)	8 (14.0)	
3	6 (13.6)	12 (21.1)	
4	6 (13.6)	11 (19.3)	
5 (least privileged)	2 (4.5)	2 (3.5)	
Unknown	7 (15.9)	3 (5.3)	
Rural – no. (%) ^{**}	5 (11.4)	6 (10.5)	0.89
Prior 6-month hospitalization – no. (%)	5 (11.4)	3 (5.3)	0.26
Key comorbidities – no. (%)			
Sleep apnea	6 (13.6)	10 (17.5)	0.59
Diabetes	10 (22.7)	8 (14.0)	0.26
CAD	6 (13.6)	6 (10.5)	0.63
CKD	3 (6.8)	3 (5.3)	0.74
COPD	1 (2.3)	1 (1.8)	0.85
Stroke	4 (9.1)	2 (3.5)	0.24
Dementia ^{††}	2 (6.7)	4 (10.0)	0.62
Heart Failure	1 (2.3)	1 (1.8)	0.85
Hip fracture	0 (0.0)	1 (1.8)	0.38
Dialysis	0 (0.0)	0 (0.0)	N/A

Number of medications – no. (%)			
Blood pressure-lowering			0.32
1	23 (52.3)	29 (50.9)	
2	12 (27.3)	18 (31.6)	
3	9 (20.5)	7 (12.3)	
≥4	0 (0.0)	3 (5.3)	
Non-blood pressure-lowering			0.32
0	5 (11.4)	11 (19.3)	
1	17 (38.6)	14 (24.6)	
2	6 (13.6)	12 (21.1)	
3	5 (11.4)	10 (17.5)	
≥4	11 (25.0)	10 (17.5)	
Total prescription (only meds taken every day)			0.93
1	4 (9.1)	7 (12.3)	
2	11 (25.0)	13 (22.8)	
3	5 (11.4)	10 (17.5)	
4	8 (18.2)	8 (14.0)	
5	6 (13.6)	6 (10.5)	
≥6	10 (22.7)	13 (22.8)	
Over the counter non-prescription			0.31
0	30 (68.2)	33 (57.9)	
1	8 (18.2)	18 (31.6)	
2	6 (13.6)	6 (10.5)	
3	0 (0.0)	0 (0.0)	
≥4	0 (0.0)	0 (0.0)	
Vitamins/herbal supplements			0.24
0	13 (29.5)	16 (28.1)	
1	12 (27.3)	15 (26.3)	
2	12 (27.3)	10 (17.5)	
3	2 (4.5)	11 (19.3)	
≥4	5 (11.4)	5 (8.8)	
Type of BP-lowering med – no. (%)			
ACE inhibitor	18 (40.9)	21 (36.8)	0.68
Angiotensin II receptor blocker	13 (29.5)	11 (19.3)	0.23
Calcium channel blocker	14 (31.8)	19 (33.3)	0.87
Diuretic	9 (20.5)	16 (28.1)	0.34
Combination BP med	9 (20.5)	16 (28.1)	0.38
Beta-blocker	8 (18.2)	14 (24.6)	0.44
Other class of BP med	2 (4.5)	0 (0.0)	0.10

Polypharmacy – no. (%) ^{††}	16 (36.4)	19 (33.3)	0.75
Resistant hypertension – no. (%) ^{§§}	9 (20.5)	10 (17.5)	0.71
* Percentages may not total 100 if data was missing. IQR denotes interquartile range. † “How many days in the past week have you exercised for 30 min or more, vigorously enough to raise your breathing rate?” Participant considered sedentary if answer = 0. ‡ As determined by the physical frailty subscale of the Tilburg questionnaire. Subscale score ≥ 3 = physically frail. § Self-reported quality of life via EQ5D5L. Higher numbers indicate higher quality of life. Range 0 – 100. A weighted, six-item test of cognitive impairment. Scores range from 0 – 28. A score of 5 – 9 suggests questionable impairment and scores of ≥ 10 suggest impairment consistent with dementia. ¶ Postal code derived social deprivation index (Pampalon) ** Postal code derived (as per Canada Post designation) †† Derived exclusively from administrative health data – hence unavailable for participants from Saskatchewan and Ontario (% calculated using only those for whom there was data) ‡‡ Taking ≥ 5 daily prescription medications. §§ Taking ≥ 3 daily BP-lowering medications.			

eTable 6. Medication use at 6-months

	Morning (n = 1680)	Bedtime (n = 1677)
Excluded participants – no. (%)	113/1680 (6.7)	159/1677 (9.5)
0 BP meds	6/113 (5.3)	11/159 (6.9)
No month 6 interview data *	107/113 (94.7)	148/159 (93.1)
≥ 1 once-daily BP med at the allocated time – no. (%)	1514/1567 (96.6)	1341/1518 (88.3)
Proportion of antihypertensive doses at the allocated time [†]	94.2%	83.5%
Number of Medications – no. (%)		
0 BP meds	6/1573 (0.4)	11/1529 (0.7)
1 BP meds	849/1573 (53.9)	789/1529 (51.6)
2 BP meds	533/1573 (33.9)	558/1529 (36.5)
3 BP meds	160/1573 (10.2)	132/1529 (8.6)
4 BP meds	22/1573 (1.4)	30/1529 (2)
5+ BP meds	3/1573 (0.2)	9/1529 (0.6)
Number of medications per participant – mean (SD)	1.589 (0.75)	1.613 (0.79)
Medication class and timing – no. (%)		
ACEI	585	529
As allocated	529/585 (90.4)	451/529 (85.3)
Off allocation	25/585 (4.3)	53/529 (10)
Twice or more daily	31/585 (5.3)	25/529 (4.7)
ARB	446	501
As allocated	425/446 (95.3)	433/501 (86.4)
Off allocation	18/446 (4)	52/501 (10.4)
Twice or more daily	3/446 (0.7)	16/501 (3.2)
Beta-blocker	268	261
As allocated	169/268 (63.1)	117/261 (44.8)
Off allocation	10/268 (3.7)	39/261 (15)
Twice or more daily	89/268 (33.2)	105/261 (40.2)
CCB	451	442
As allocated	406/451 (90)	391/442 (88.5)
Off allocation	31/451 (6.9)	44/442 (9.9)
Twice or more daily	14/451 (3.1)	7/442 (1.6)
Combination BP med	282	285
As allocated	272/282 (96.5)	225/285 (78.9)
Off allocation	9/282 (3.2)	58/285 (20.4)

	Morning (n = 1680)	Bedtime (n = 1677)
Twice or more daily	1/282 (0.4)	2/285 (0.7)
Diuretic	450	424
As allocated	439/450 (97.6)	277/424 (65.3)
Off allocation	9/450 (2)	139/424 (32.8)
Twice or more daily	2/450 (0.4)	8/424 (1.9)
Other	17	25
As allocated	5/17 (29.4)	15/25 (60)
Off allocation	8/17 (47.1)	1/25 (4)
Twice or more daily	4/17 (23.5)	9/25 (36)
* Participants not providing data had dropped out, or missed the 6-month interview. ‡ Proportion of doses at the allocated time incorporates medications dosed multiple times per day by considering them as ½ dose in the AM and ½ dose in the PM.		

eTable 7. Compliance with allocation time over the duration of the trial

Follow-up interview date	Allocation	>= 1 once-daily BP med at allocated time	Proportion of doses at allocated time*	Medication administration time – no. of Rx (%)				
				Morning	Noon	Dinner	Bedtime	Takes twice or more
Month 6	Morning n = 1567; Rx = 2499	1514/1567 (96.6%)	94.2%	2231/2499 (89%)	14/2499 (0.6%)	17/2499 (0.7%)	93/2499 (4%)	144/2499 (5.8%)
	Bedtime n = 1518; Rx = 2467	1341/1518 (88.3%)	83.5%	375/2467 (15.2%)	11/2467 (0.4%)	93/2467 (3.8%)	1816/2467 (73.6%)	172/2467 (7%)
Month 12	Morning n = 1532; Rx = 2455	1461/1532 (95.4%)	92.5%	2135/2455 (87%)	20/2455 (0.8%)	34/2455 (1.4%)	110/2455 (4.5%)	156/2455 (6.3%)
	Bedtime n = 1483; Rx = 2411	1257/1483 (84.8%)	79.6%	459/2411 (19%)	12/2411 (0.5%)	110/2411 (4.6%)	1667/2411 (69.1%)	163/2411 (6.8%)
Month 18	Morning n = 1484; Rx = 2398	1395/1484 (94.0%)	90.8%	2035/2398 (84.9%)	25/2398 (1%)	40/2398 (1.7%)	139/2398 (5.8%)	159/2398 (6.6%)
	Bedtime n = 1440; Rx = 2372	1179/1440 (81.9%)	76.6%	521/2372 (22%)	17/2372 (0.7%)	105/2372 (4.4%)	1557/2372 (66.6%)	172/2372 (7.3%)
Month 24	Morning n = 1385; Rx = 2223	1294/1385 (93.4%)	90%	1876/2223 (84.4%)	17/2223 (0.8%)	35/2223 (1.6%)	143/2223 (6.4%)	152/2223 (6.8%)
	Bedtime n = 1347; Rx = 2235	1083/1347 (80.4%)	74.5%	529/2235 (23.7%)	19/2235 (0.8%)	100/2235 (4.5%)	1412/2235 (63.2%)	175/2235 (7.8%)
Month 30	Morning n = 1229; Rx = 1975	1138/1229 (92.6%)	89.9%	1645/1975 (83.3%)	18/1975 (0.9%)	32/1975 (1.6%)	131/1975 (6.6%)	149/1975 (7.6%)
	Bedtime n = 1201; Rx = 1993	934/1201 (77.8%)	72.4%	502/1993 (25.2%)	18/1993 (0.9%)	92/1993 (4.6%)	1215/1993 (61%)	166/1993 (8.3%)
Month 36	Morning n = 1076; Rx = 1755	987/1076 (91.7%)	88.3%	1433/1755 (81.6%)	13/1755 (0.7%)	33/1755 (1.9%)	131/1755 (7.5%)	145/1755 (8.3%)
	Bedtime n = 1053; Rx = 1771	820/1053 (77.9%)	72.5%	441/1771 (24.9%)	14/1771 (0.8%)	91/1771 (5.1%)	1063/1771 (60%)	162/1771 (9.2%)
Month 42	Morning n = 1012; Rx = 1662	932/1012 (92.1%)	88.3%	1352/1662 (81.3%)	15/1662 (0.9%)	31/1662 (1.9%)	129/1662 (7.8%)	135/1662 (8.1%)
	Bedtime n = 987; Rx = 1673	769/987 (77.9%)	71.4%	438/1673 (26.2%)	18/1673 (1%)	78/1673 (4.7%)	1003/1673 (60%)	136/1673 (8.1%)
Month 48	Morning n = 840; Rx = 1393	760/840 (90.5%)	86.3%	1088/1393 (78.1%)	17/1393 (1.2%)	27/1393 (1.9%)	132/1393 (9.5%)	129/1393 (9.3%)
	Bedtime n = 831; Rx = 1405	617/831 (74.2%)	69%	404/1405 (28.8%)	16/1405 (1.1%)	66/1405 (4.7%)	793/1405 (56.4%)	126/1405 (9%)

Month 54	Morning n = 736; Rx = 1227	675/736 (91.7%)	87%	973/1227 (79.3%)	12/1227 (1%)	21/1227 (1.7%)	116/1227 (9.4%)	105/1227 (8.6%)
	Bedtime n = 720; Rx = 1216	525/720 (72.9%)	67.9%	363/1216 (29.8%)	12/1216 (1%)	58/1216 (4.8%)	676/1216 (55.6%)	107/1216 (8.8%)
Month 60	Morning n = 537; Rx = 881	486/537 (90.5%)	85.7%	693/881 (78.7%)	3/881 (0.3%)	19/881 (2.2%)	98/881 (11.1%)	68/881 (7.7%)
	Bedtime n = 508; Rx = 860	379/508 (74.6%)	69.1%	251/860 (29.2%)	7/860 (0.8%)	41/860 (4.8%)	479/860 (55.7%)	82/860 (9.5%)
Month 66	Morning n = 321; Rx = 527	292/321 (91.0%)	85.8%	419/527 (79.5%)	5/527 (0.9%)	4/527 (0.8%)	62/527 (11.8%)	37/527 (7%)
	Bedtime n = 298; Rx = 505	221/298 (74.2%)	68.7%	145/505 (28.7%)	4/505 (0.8%)	25/505 (5%)	284/505 (56.2%)	47/505 (9.3%)
Month 72	Morning n = 190; Rx = 309	167/190 (87.9%)	83.7%	235/309 (76.1%)	4/309 (1.3%)	2/309 (0.6%)	38/309 (12.3%)	30/309 (9.7%)
	Bedtime n = 169; Rx = 288	118/169 (69.8%)	65.5%	89/288 (30.9%)	4/288 (1.4%)	15/288 (5.2%)	155/288 (53.8%)	25/288 (8.7%)

* Calculated using the average percentage of medications taken at the allocated time for all participants. Morning group medications were considered to be taken as allocated if used in the morning or at lunch-time. Bedtime group medications were considered to be taken as allocated if used at bedtime or at dinner. Medications taken two or more times daily were considered as ½ dose in the morning and ½ dose in the evening for this calculation.

eTable 8. Success of once-daily medication changes at 6-months, along with reasons for failure, by medication class

	ACEI		ARB		Beta-Blocker		CCB	
	AM	PM	AM	PM	AM	PM	AM	PM
Once daily baseline medications still in use at 6-months*	510	461	393	440	138	127	398	392
Once Daily Meds at Baseline Indicated for Change	90/510 (17.6%)	373/461 (80.9%)	60/393 (15.3%)	361/440 (82.0%)	33/138 (23.9%)	96/127 (75.6%)	106/398 (26.6%)	280/392 (71.4%)
Successful Timing Change	74/90 (82.2%)	324/373 (86.9%)	54/60 (90.0%)	316/361 (87.5%)	26/33 (78.8%)	66/96 (68.8%)	89/106 (84.0%)	244/280 (87.1%)
Unsuccessful Timing Change	16/90 (17.8%)	49/373 (13.1%)	6/60 (10.0%)	45/361 (12.5%)	7/33 (21.2%)	30/96 (31.3%)	17/106 (16.0%)	36/280 (12.9%)
No Attempts at 6 months	4/16 (25.0%)	11/49 (22.4%)	2/6 (33.3%)	8/45 (17.8%)	4/7 (57.1%)	18/30 (60.0%)	8/17 (47.1%)	10/36 (27.8%)
Failed Attempts at 6 months [†]	12/16 (75.0%)	38/49 (77.6%)	4/6 (66.7%)	37/45 (82.2%)	3/7 (42.9%)	12/30 (40.0%)	9/17 (52.9%)	26/36 (72.2%)
BP Not Controlled	2/12 (16.7%)	6/38 (15.8%)		9/37 (24.3%)		3/12 (25.0%)	2/9 (22.2%)	6/26 (23.1%)
Medical Professional's Advice	3/12 (25.0%)	4/38 (10.5%)		5/37 (13.5%)	3/3 (100%)	1/12 (8.3%)	4/9 (44.4%)	1/26 (3.8%)
Inconvenient		6/38 (15.8%)	1/4 (25.0%)	7/37 (18.9%)		3/12 (25.0%)	1/9 (11.1%)	3/26 (11.5%)
Kept Forgetting	1/12 (8.3%)	13/38 (34.2%)	3/4 (75.0%)	7/37 (18.9%)		2/12 (16.7%)	1/9 (11.1%)	7/26 (26.9%)
Believed bedtime improves health outcomes								
Concern about taking too many medications at the same time								
Side Effects	8/12 (66.7%)	13/38 (34.2%)	2/4 (50.0%)	13/37 (35.1%)	1/3 (33.3%)	4/12 (33.3%)	5/9 (55.6%)	10/26 (38.5%)
Ankle Swelling		1/13 (7.7%)						
Coughing							1/5 (20.0%)	
Dizziness/pre-syncope/syncope	6/8 (75.0%)	2/13 (15.4%)					1/5 (20.0%)	2/10 (20.0%)

	ACEI		ARB		Beta-Blocker		CCB	
	AM	PM	AM	PM	AM	PM	AM	PM
Dry Mouth		2/13 (15.4%)						
Erectile Dysfunction		1/13 (7.7%)						
Falling		1/13 (7.7%)						
Urination at Night		5/13 (38.5%)		6/13 (46.2%)		4/4 (100%)		4/10 (40.0%)
Weakness/Fatigue	3/8 (37.5%)		2/2 (100%)		1/1 (100%)		1/5 (20.0%)	
Headache		1/13 (7.7%)					1/5 (20.0%)	2/10 (20.0%)
Arrhythmias				2/13 (15.4%)				
General Malaise				2/13 (15.4%)				1/10 (10.0%)
Sleep disturbance - Insomnia		2/13 (15.4%)		1/13 (7.7%)				1/10 (10.0%)
Sleep disturbance - Night sweats		1/13 (7.7%)					1/5 (20.0%)	
Sleep disturbance - Nightmares		1/13 (7.7%)						
Sleep disturbance - Leg cramps								
Sleep disturbance - Leg aches		1/13 (7.7%)						
Other Side Effects		1/13 (7.7%)		2/13 (15.4%)				
Other Reason/Cannot Remember		2/38 (5.3%)		1/37 (7.7%)				2/26 (7.7%)

	Combination medications		Diuretic		Other	
	AM	PM	AM	PM	AM	PM
Once daily baseline medications still in use at 6 Months*	263	267	409	370	11	12
Once Daily Meds at Baseline Indicated for Change	16/263 (6.1%)	249/267 (93.3%)	25/409 (6.1%)	352/370 (95.1%)	8/11 (72.7%)	5/12 (41.7%)
Successful Timing Change	12/16 (75.0%)	194/249 (77.9%)	20/25 (80.0%)	238/352 (67.6%)	1/8 (12.5%)	4/5 (80.0%)
Unsuccessful Timing Change	4/16 (25.0%)	55/249 (22.1%)	5/25 (20.0%)	114/352 (32.4%)	7/8 (87.5%)	1/5 (20.0%)
No Attempts at 6 months	2/4 (50.0%)	7/55 (12.7%)	4/5 (80.0%)	55/114 (48.2%)	5/7 (71.4%)	1/1 (100%)
Failed Attempts at 6 months [†]	2/4 (50.0%)	48/55 (87.3%)	1/5 (20.0%)	59/114 (51.8%)	2/7 (28.6%)	
BP Not Controlled	1/2 (50.0%)	5/48 (10.4%)	1/1 (100%)		1/2 (50.0%)	
Medical Professional's Advice		9/48 (18.8%)		11/59 (18.6%)	1/2 (50.0%)	
Inconvenient	1/2 (50.0%)	4/48 (8.3%)		7/59 (11.9%)		
Kept Forgetting		6/48 (12.5%)		13/59 (22.0%)		
Believed bedtime improves health outcomes						
Concern about taking too many medications at the same time					1/2 (50.0%)	
Side Effects		30/48 (62.5%)		33/59 (55.9%)		
Ankle Swelling						
Coughing						
Dizziness/pre-syncope/syncope		1/30 (3.3%)				
Dry Mouth						

	Combination medications		Diuretic		Other	
	AM	PM	AM	PM	AM	PM
Erectile Dysfunction		1/30 (3.3%)				
Falling						
Urination at Night		25/30 (83.3%)		29/33 (87.9%)		
Weakness/Fatigue		1/30 (3.3%)		1/33 (3.0%)		
Headache		2/30 (6.7%)		2/33 (6.1%)		
Arrhythmias						
General Malaise						
Sleep disturbance - Insomnia		1/30 (3.3%)		1/33 (3.0%)		
Sleep disturbance - Night sweats						
Sleep disturbance - Nightmares						
Sleep disturbance - Leg cramps		1/30 (3.3%)				
Sleep disturbance - Leg aches						
Other Side Effects		2/30 (6.7%)		1/33 (3.0%)		
Other Reason/Cannot Remember	1/2 (50.0%)					
* Table reports only once-daily medications that were present at baseline and still in use at 6-months. Medications discontinued, or newly added post randomization, are excluded. † The same participant can select multiple reasons for why they failed to make a timing change. Consequently, they may appear in more than one category.						

eTable 9. 24-hour ambulatory blood pressure monitoring results

Blood Pressure	Treatment (Bedtime) N = 151	Control (Morning) N = 151	Mean Difference (95% CI)	p-value
Mean Awake SBP* - mmHg (SD)	133.8 ± 13.0	136.2 ± 15.6	-2.4 (-5.7 to 0.9)	0.15
Mean Awake DBP* - mmHg (SD)	75.2 ± 8.9	75.6 ± 10.0	-0.4 (-2.5 to 1.8)	0.72
Mean Sleep-time SBP* - mmHg (SD)	116.5 ± 14.9	123.9 ± 18.2	-7.4 (-11.2 to -3.7)	<0.001
Mean Sleep-time DBP* - mmHg (SD)	62.9 ± 9.2	65.5 ± 10.6	-2.7 (-4.9 to -0.4)	0.02
Sleep-time SBP* dip [†] - % (SD)	12.9 ± 8.3	9.1 ± 8.4	3.8 (1.9 to 5.7)	<0.001
Non-dipper [†] - No. (%)	54 (35.8)	78 (51.7)	-	0.005
Appropriate dipper [†] - No. (%)	70 (46.4)	58 (38.4)	-	0.16
Extreme dipper [†] - No. (%)	27 (17.9)	15 (9.9)	-	0.05
Awake Blood Pressure Controlled (BP* < 130/80 mmHg) - No. (%)	62 (41.1)	49 (32.5)	-	0.12
Awake SBP* Controlled (SBP* < 130 mmHg) - No. (%)	66 (43.7)	53 (35.1)	-	0.13
Awake DBP* Controlled (DBP* < 80 mmHg) - No. (%)	111 (73.5)	102 (67.5)	-	0.25
*BP = Blood pressure, SBP = Systolic blood pressure, DBP = Diastolic blood pressure † Asleep SBP dip is the % lowering of sleep-time SBP compared to daytime SBP. Non-dipper = sleep-time SBP dip < 10%; Appropriate dipper = sleep-time SBP dip between 10 and 20%; Extreme dipper = sleep-time ABP dip > 20%				